

DR. ANES`S BASICS of FOOD-DRUG INTERACTIONS

For Therapeutic nutrition BC students

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Main Reference

Joseph I. Boullata and Vincent T. Armenti. **Handbook of drug-nutrient interactions**. 2nd Ed. 2010, Humana Press, Springer, USA

CHAPTER I: INTRODUCTION TO FOOD-DRUG INTERACTIONS

1.1. Definitions

- **Drug (medicine)** is defined as a substance used in the diagnosis, treatment, or prevention of a disease. It is also the active ingredient in a given medication.
- **Medication (drug product)** is the product containing the drug (medicine) that is given or administered to the patient.
- **Food** A substance that is eaten, drunk or taken into the body to provide nutritional support and sustain life.

- **Meal**

A part of food served at a given time daily (breakfast, lunch and dinner).

- **Nutrient**

- ✓ It is the **valuable** component in food that is essential: to provide energy, to build the body or regulate chemical process in the body.
- ✓ Types of nutrients include (1) **Macronutrients:** Carbohydrates, Lipids, Proteins, (2) **Micronutrients :** vitamins , minerals and trace elements daily allowance less than 100 mg) and (3) **Water.**

- **Dietary supplement**

A product taken orally that contains one or more ingredients (e.g. amino acid, vitamins, minerals, etc.) that are intended to improve quality of a person`s diet but are not considered food. Example : multivitamin capsules.

- **Non-nutritive dietary**

Nonnutritive dietary include all what we eat, drink or inhale but don't contain any nutrient. They are generally used to improve mood (euphoria), physical and mental activity. Examples: **coffee , tea, alcohol and tobacco smoke.**

- **Nutritional status**

- ✓ Condition of an individual's health influenced by the intake/deficiency of nutrients.
- ✓ Types:
 - (i) Normal
 - (ii) Abnormal
 - Hypo nutrition status : e.g. malnutrition, hypocalcaemia
 - Over nutritive status: e.g. obesity , hyperglycemia

- **Drug-disposition (pharmacokinetics)**

Once a drug is administered or taken by the patient, it undergoes a series of process in the body including (absorption, distribution, Elimination (metabolism + excretion). All these processes are collectively called “**Drug disposition**” or “**Pharmacokinetics**”.

- **Drug effect (Pharmacological response)**

- It is the physiological change that occurs in the body due to taking of the drug
- Types
 - (i) **Desired effects (Drug efficacy)** for which the drug is taken
 - (ii) **Undesired effects.**
 - Simple and tolerated : Side effects
 - Serious : adverse effects
 - Very serious: exacerbation of adverse effects (i.e. **Drug toxicity**)

Example

Isoniazid (anti-TB)

Desired effect : elimination of *Mycobacterium tuberculosis* from the body

Undesired :

- Simple : dizziness

- Adverse : neuralgia
- Very serious (Drug toxicity): Peripheral neuritis

1.2. Food-drug Interaction

1.2.1. Definition

It occurs when taking food (or deficiency in a nutrient) (or abnormal nutrition status) cause Change (**increase or decrease**) in normal “ **drug disposition**” or “ **drug effect**”. Or occurs when a drug change normal nutrition status in the body

Causes

- (i) Taking of a food with a drug (Food + drug)
- (ii) Inappropriate nutritional status while taking a drug (Nutrition status + drug)

1.2.2. Classification of Food-drug interaction (FDI)

Type	Precipitator (Cause)	Object (affected)	Outcome
Food affecting Drug	Intake or deficiency of Food, nutrient , dietary supplement Or change in nutritional Status in body	Drug	Failure of drug therapy Or Drug toxicity
Drug affecting Food	Drug	Nutrients	Change in nutritional status

1.2.3. FDI .. Is it BAD (negative) or GOOD (positive) ?

Type of FDI	If FDI cause
BAD (Negative)	Decrease drug effect Or Increase drug toxicity
GOOD (Positive)	increase drug effect (but not drug toxicity) Or decrease drug toxicity

1.2.4. History of FDI

Year	Precipitator	Object	Clinical Outcome
1941	Vitamin C	Barbiturates (e.g. phenobarbitone): (antiepileptic and hypnotic drug)	<u>Food affecting drug</u> Deficiency in Vitamin C increase excretion of the antiepileptic Failure of antiepileptic therapy
1954	Isoniazide (anti-TB)	vitamin B6	<u>Drug affecting food</u> (increase the metabolism of the vitamin) ... peripheral neuritis

1.2.5. FDI in specific populations of patients

Specific populations of patients are patients who have serious conditions/disorders and need specific care. These patients are more affected by drugs, nutritional status and drug interactions than other group of patients. Extra-Care is required while assessment of FDI in these groups

Specific populations of patients include:

- Patients on extremities of life (Children - Elderly)
- Pregnant and lactating women
- Cancer patients
- Patients with immunity-related disorders
- Patients with serious chronic diseases: Renal failure, hepatic failure, Diabetes, CVS patients, Status asthmaticus, serious infections (TB, AIDS, etc.)

1.2.6. Assessment “Evaluation” of FDI

Clinical outcome	Severity of interaction
Sever e.g. life-threatening, complete failure of drug therapy, sever change in nutritional status	Major
Clinical outcome is present not , but not sever	Moderate
Clinical outcome is very slight Or Unlikely (doubtful) or data is (insufficient)	Minor

What should we do ?

- Major FDI : Stop the food or nutrient that cause the interaction
- Moderate FDI : Adjust food regimen
- Minor FDI : Nothing but keep monitoring the case.

FDA Probability of a new/unknown FDI

Table 4

The Drug Interaction Probability Scoring System and Scale (132)

<i>Question</i>	<i>Reply</i>		
	<i>Yes</i>	<i>Unknown or N/A</i>	<i>No</i>
• Are there previous <i>credible</i> reports of this interaction in humans?	+ 1	0	−1
• Is the observed interaction consistent with the known interactive properties of the precipitating factor?	+ 1	0	−1
• Is the observed interaction consistent with the known interactive properties of the object?	+ 1	0	−1
• Is the event consistent with the known or reasonable time course of the interaction (onset and/or offset)?	+ 1	0	−1
• Did the interaction remit upon dechallenge of the precipitating factor with no change in the object?	+ 1	0	−2
○ If so, did the interaction reappear when the precipitating factor was readministered in the presence of continued use of the object?	+ 2	0	−1
• Are there reasonable alternative causes for the event? [†]	−1	0	+ 1
• Was the object of the interaction detectable in the blood or other fluids in concentrations consistent with the proposed interaction?	+ 1	0	0
• Was the drug interaction confirmed by any objective evidence consistent with the effects on the object (other than concentrations from the previous question)?	+ 1	0	0
• Was the interaction greater when the precipitating factor was increased or less when the precipitating factor was decreased?	+ 1	0	−1
Total score ____			
Score		Probability	
<2		Doubtful	
2–4		Possible	
5–8		Probable	
>8		Highly probable	

CHAPTER II: INFLUENCE OF FOOD ON DRUG DISPOSITION/EFFECT

2.1. Influence of Food on drug absorption

❖ Pre-absorption processes

Prior to absorption, the drug must undergo a series of processes to make the drug available for absorption. These processes are

- (i) **Drug release:** liberation of drug (as free particles) form dosage form.
It occurs at the surface of stomach fluid and its rate varies according to type of dosage form

“ Liquid (solution, suspension, emulsion) > Solids (capsules, tablets)”

- (ii) **Drug dissolution**

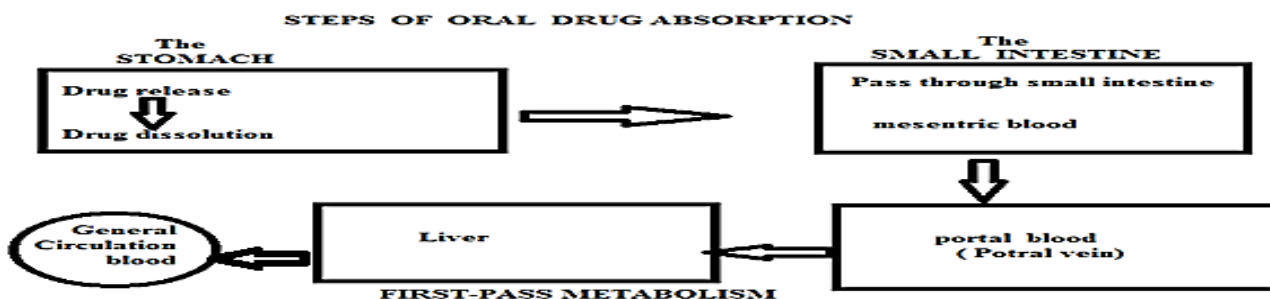
It occurs at the bulk of gastric fluid. In this process, free drug particles are reduced to Nano-particles and is ready for absorption

Failure of drug release or drug dissolution will result in failure of drug absorption

❖ Drug absorption

It is the process of drug transport from the site of absorption (e.g. small intestine) to general blood circulation.

Steps of oral Drug absorption



First-pass to liver

- ✓ After drug absorption through small intestine, The drug reaches specific blood vessels called “mesenteric blood vessels” which drains in the portal vein that drains the blood into gall bladder and then liver before reaching general circulation as follows:

“Small intestine Mesenteric blood Portal vein Gall bladder ... liver General circulation to heart”

- ✓ The passage of drug to liver before reaching general circulation is called “First-pass” . More discussion about first-pass metabolism
- ✓ In the liver, some drugs e.g. propranolol, lidocaine, morphine undergoes to metabolism which may destroy most of the drug dose.

Note : Most drugs undergo minor or no metabolism during First-pass

Mechanisms of drug absorption

The process by which the drug pass through membrane of small intestine occurs for most drugs spontaneously **by a mechanism called “passive diffusion**; It needs no energy or carrier. However, some drugs are absorbed by “**Active transport**” mechanism. The mechanism needs energy and specific protein transporters (carriers)

Example of important absorption carriers

Carrier (Transporter)	Example of drugs that need the transporter
Amino acid transporter (AAT)	Levodopa, methyldopa
Oligopeptide transporter (OPT)	Captopril, lisinopril Cephalosporins: cefadroxil, cephradine, cefixime
Monocarboxylic acid transporter (MAT)	pravastatin
Organic anion transporter polypeptide (OATPs)	Levothyroxine, cyclosporine, acebutolol , atenolol, fexofenadine , ciprofloxacin , norfloxacin

Mechanisms of Food interaction on drug absorption

(i) Food decreases drug absorption

Absorption of many drugs decreases or delayed if taken with food. These drugs should be taken on an empty stomach (at least 1 hour before meal) (or 2hours after meal).

Mechanisms of food decreasing drug absorption are as follow:

- ✓ **Complexation or chelation:** of food or nutrient with drug
e.g. Calcium in milk chelate tetracycline, Iron with ciprofloxacin
- ✓ **Competition or inhibition of carriers :** Amino-acids present in food may compete with some drug “ e.g. levodopa” for specific absorption transporter (amino acid transporter) so reducing absorption of the drug. Another example is Grapefruit, apple juice and orange juice which inhibit specific transporter carrier (OATP; organic anion transporter protein) thereby decreasing the absorption of many drugs e.g. fexofenadine, atenolol and cyclosporine
- ✓ Presence of food in the stomach (full stomach) generally slow down gastric emptying time and hence delays absorption of drugs which occurs mostly in the small intestine (duodenum).

Drugs should be taken on an empty stomach

Type	Drug	Clinical outcomes if food is taken	
CNS drugs	Phenytoin (antiepileptic)	Food decreases absorption	Decrease or loss of drug efficacy
	Zolpidem (Hypnotic/sedative)	Food delay absorption	
	Levodopa (anti-parkinsonism)	Food reduces absorption especially food containing proteins (amino acids) that compete with the drug for carrier (amino acid transporter AAT)	
	Acetaminophen (analgesic)	Food rich in fiber (e.g. pectin) decrease the drug absorption	
Drugs for bronchial asthma	Theophylline	Food decrease absorption	
	Zafirlukast		
CVS drugs and drugs affecting blood	Digoxin (cardiac stimulant)	Food delays absorption and may decrease peak concentrations	
	Furosemide (Diuretic)	Absorption is increased in the fasting state	
	Atenolol, acebutolol	Grapefruit or orange juice which inhibit OATP thereby decreasing drug absorption	
	Diltiazem (Antihypertensive ; CCB)	Absorption is increased in the fasting state	
	Warfarin (oral anticoagulant)	Food decrease absorption	
	Aliskiren (antihypertensive; Renin inhibitor)		
Antihistamines	Fexofenadine	Grapefruit, apple juice and orange juice inhibit OATP and thereby decreasing the drug absorption	

Drugs should be taken on an empty stomach

Antibacterials	Azithromycin	Food decreases absorption of capsules (but not tablets or suspensions) by 50%	Decrease or loss of drug efficacy
	Tetracycline, minocycline, doxycycline	Food , in general, especially food containing divalent or trivalent minerals (e.g. Calcium, iron, etc)	
	Ciprofloxacin, norfloxacin	Food , in general, especially food containing divalent or trivalent minerals (e.g. Calcium, iron, etc.) Ciprofloxacin absorption also decreases with orange juice which inhibit OATP and thereby decreasing drug absorption	
	Ampicillin	Food decrease absorption	
	Metronidazole	Food delay absorption	
Antiviral	Didanosine (anti viral HIV)	Food decrease or delay absorption	
Oral hypoglycemic	Glipizide		
Thyroid hormone	Levothyroxine	Food in general decrease oral absorption especially grapefruit which inhibits OATP thereby decreasing drug absorption	
Immunosuppressant	Cyclosporine	Apple juice inhibits OATP thereby decreasing drug absorption	
GIT drugs (for peptic ulcer)	Omeprazole, lansoprazole, esomeprazole, Rabeprazole, pantoprazole	Food decrease and delay absorption	

(ii) Food increases drug absorption

In contrast to previously mentioned, food may improve absorption of certain drugs or food may decrease GIT side effect. These drugs are better to be used with food.

Mechanism of food improving absorption of some drugs may be:

- ✓ Presence of food in the stomach (full stomach) slow down gastric emptying rate and hence make drugs stay for a longer time in the stomach. This is beneficial for drugs that have low dissolution in the stomach (lipophilic drugs) and hence such dissolution will be improved.
- ✓ Presence of food in the stomach (full stomach) stimulate the secretion of bile salts (from gall bladder). Bile salts are surfactant and can enhance dissolution of lipophilic drugs and hence improve their absorption
- ✓ Presence of food in the stomach (full stomach) reduces the direct irritation of stomach and GIT caused by some drugs. Direct irritation may cause GIT upset and increase GIT motility which can reduce drug absorption.
- ✓ Some food may inhibit "First-pass metabolism".

Drugs should be taken with food

Type	Drug	Clinical outcome if food is taken	
CNS drugs	Carbamazepine	Food decreases	Decrease Drug toxicity
CVS drugs	Carvedilol, metoprolol	GIT upset	
NSAIDs analgesic-antiinflammatory	Aspirin, diclofenac, ibuprofen, celecoxib, meloxicam, piroxicam , ketoprofen, etc.	Food decreases GIT upset	
Corticosteroids anti-inflammatory	Prednisolone, methylprednisolone	Food decreases GIT upset	
Oral hypoglycemic	metformin		
Type	Drug	Clinical outcome if food is taken	
Antibacterials	Amoxicillin + calvulenate	Food increase absorption of and reduces GIT upset	Increase drug efficacy
Antimalarial	Atovaquone	Food (high fat)	
Antifungal	Ketoconazole	increase absorption	
Minerals	Potassium chloride	Food decrease GIT upset	
Vitamins	Cyanocobalamin (Vit B12)	Food increase absorption	
	Riboflavin B2		

(iii) Effect of food on Gastric emptying time (GET)

- GET : time required for the stomach to empty its content.
- Normal GET is approximately 2 hours after meal

Food that can slow down GET :

- Full stomach and Heavy meal
- Large volume of drinks
- Fat-rich food , High-protein with low fat
- High viscous (e.g. Fiber-containing food)

- Hot drinks or Ice or very cold drinks
- Acidic food (e.g. citrus fruits)
- **Food that can accelerate GET :**
 - Empty stomach (fasting)
 - Liquids (normal temperature, low viscous)

What happens to drug absorption if GET slow down ?

- ✓ Generally, it delays absorption of most drugs
- ✓ Drugs that should not stay for long time in the stomach, because they are acid labile (destroyed by gastric pH), will have low absorption extent e.g. digoxin, ampicillin
- ✓ Few drugs may have greater absorption e.g. Riboflavin (vitamin B2), cyanocobalmine (vitamin B12), ketoconazole.

2.2. Effect of Food on drug distribution

❖ Drug distribution definition

It is the process by which a drug transport from blood circulation to body tissue (i.e. to target tissue).

❖ Drug distribution process

- ✓ Following first-pass, the drug reaches general circulation directly to the heart.
- Steps in drug distribution
1. The heart pumps blood and drug to circulate in the body.
 2. The drug transport from blood to extracellular fluid and then to intercellular fluid near to target cell. The drug should be free (unbound) to anything to can transport from the blood to extracellular fluid.

3. The target of some drugs (receptor, enzyme, etc.) may be located externally on the membrane of the target cell so the drug can binds easily to it .

However, the target of some drugs (receptor, enzyme, etc.) may be inside the cell. So the drug must pass through the cell membrane to reach it. Some drugs pass spontaneously (passive diffusion) but other drugs may need specific proteins (carriers; transporter) to pass the cell membrane.

“ Blood circulation HeartBlood circulation extracellular fluid intercellular fluid cellular target (on cell membrane; inside the cell)..... drug back to blood”

❖ **Factors affecting drug distribution**

- ✓ Most drugs have wide distribution (great V_D) and so can reach most body tissues.
- ✓ However, some drugs may have narrow distribution (small V_D) due to many reasons :
 - **Heart condition** : it affects cardiac output and blood flow to target tissue. If the C.O.P. or blood flow are small, the drug distribution is narrow (small V_D)
 - **Plasma-protein binding (PPB):** some drugs (e.g. furosemide) may be extensively , but reversibly, bound to plasma proteins (albumin, alpha-glycoprotein, lipoproteins) and so they continue to circulate in blood till get free from PPB after long time. The distribution of these drugs is Narrow. In addition, these drugs also will have slow metabolism, excretion and clearance and hence long duration of action
 - **Barriers at “target tissue”** e.g. placenta, BBB. Some drugs can not penetrate these barriers and hence can not reach target tissue. Example
First-generation cephalosporins can not penetrate BBP and so can not treat “meningitis”

- Some drugs need **specific proteins (transporter; carrier)** to can reach the target inside the cell. Examples metformin need a transporter called (OCT) and atorvastatin need a carrier called (OATP)

How food can affect drug distribution ?

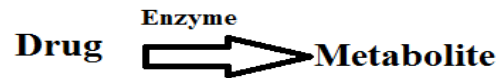
- ✓ In general, food has a little effect on distribution of most drugs.
- ✓ However, some citrus foods e.g. (Grapefruit, orange juice and apple juice) can inhibit specific transporter (e.g. OATP) which play a role in distribution of some drugs.

Distribution transporter (Carrier)	Drugs	How grapefruit, orange juice or apple juice (inhibitors of OATP) can affect drug distribution?	Clinical outcome
OATP 1A2	ciprofloxacin levofloxacin,	Decreasing distribution of those drugs to CNS thereby decreasing the drug effect.	Decrease drug efficacy
OATP 2B1	Atorvastatin, pravastatin	Decrease distribution of the drugs to liver, thereby decreasing their effect	
OATP2B1 and 1B3	Methotrexate rifampicin	Decreasing distribution to liver, thereby decreasing drug metabolism	Increase drug toxicity

2.3. Effect of Food on drug Metabolism

❖ Drug metabolism Definition

It is the process of changing the chemical structure of the drug into a **metabolite**.



❖ Functions of drug Metabolism

- ✓ To convert the drug into (polar metabolite) which can be excreted easily.
- ✓ Inactivation (ending the drug effect; most) or activation (enhancing drug effect e.g. clopidogrel)

Metabolic process	Drug	Metabolite	Example
Inactivation	Active	Inactive	Most drugs
Activation	Active	Active	Codeine, verapamil
	Inactive	Active	Clopidogrel, sulfasalazine

- ✓ Few drugs e.g. clopidogrel (inactive) and is converted by metabolism to active metabolite

❖ Metabolism organs

- ✓ Liver is the main site of drug metabolism
- ✓ Few other organs may have a role in metabolism : e.g. small intestine (first-pass metabolism) , Blood , etc.

❖ Drug-metabolizing Enzymes

Metabolism occur by reaction of the drug with an enzyme

The most important enzymes that metabolize drugs are

1. Phase I reactions : Oxidation, reduction, hydrolysis

CYPs enzymes (Cytochrome P 450 enzymes)

These are the major drug metabolizing enzymes for most drugs

- **CYP3A4** : metabolizes 45 % of drugs
- **CYP2D6**: metabolizes 25 % of drugs
- **CYP2C9, CYP2C19, CYP1A**: metabolize 24 % of drugs
- **CYP2B6, CYP2E1, CYP1A2**: metabolize 6 % (only few drugs)

2. Phase II: Reactions (Conjugation)

(i) UGTs (Uridine diphosphate GlucuronosylTransferases)

It produces metabolites as **glucuronide conjugate** . Few drugs are metabolized lonely by this enzyme. However, many many drugs are partially metabolized by this enzyme (i.e. are metabolized by CYP together with this enzyme)

(ii) GST (Glutathione S-Transferases)

It produces metabolites as **glutathione conjugate**. The enzyme is responsible only for very few drugs as follows

- Acetaminophen : The drug is mainly metabolized by UGT. If the patient suffers from hepatic insufficiency, the metabolites are further metabolized by GST to produce toxic glutathione conjugate.
- Busulfan : It metabolized mainly by GST

(iii) Others

- ✓ Sulfate conjugation
- ✓ Methyl conjugation
- ✓ Acetylation conjugation
- ✓ Peptide conjugation

How can food affect drug metabolism ?

- ✓ Food can affect drug metabolism either
 - “First-pass metabolism” : in small intestine and liver before reaching general circulation, thereby affecting drug absorption
 - General metabolism : after drug distribution, thereby affecting drug elimination
- ✓ Food can affect drug distribution to liver (see previous; effect of grapefruit, orange juice and apple juice which decrease drug distribution to liver) thereby decreasing drug metabolism
- ✓ Some food can increase drug metabolism by (Enzyme induction)

Results

- Inactivation of drug loss of efficacy if (active inactive)
- Activation of drug increase toxicity (inactive Active or active Active)
- ✓ Some food can decrease drug metabolism by (Enzyme inhibition)

Results

- Prolong Activation of drug Drug toxicity... if (active inactive)
- inactivation..... loss efficacy (inactive Active) or active Active)

Effect of diet components on theophylline metabolism

Diet	Effect on metabolism	Clinical outcome
Normal diet (protein 5-9% , fat 35 %, carbohydrates 57-60 %)	Normal metabolism	Normal drug efficacy
High protein diet + Low carbohydrate diet (protein 50% , fat 30 %, carbohydrates 20 %)	Metabolism increase	Loss of drug efficacy
High carbohydrate + low fat (protein 10% , fat 10 %, carbohydrates 80 %)	Normal metabolism (Metabolism unchanged)	Normal drug efficacy
Or High Fat + low carbohydrate (protein 10% , fat 70 %, carbohydrates 20 %)	Decrease metabolism in patients with asthma	Increase drug toxicity

Effect of some foods/nutrient on drug metabolism

Food or nutrient	Effect of food	Clinical outcome
Food / nutrient (Enzyme inducers) increase metabolism		
Cruciferous vegetables (Cabbage ملفوف, broccolis القرنبيط, cauliflower بروكولي)	CYP Enzyme inducer ... increase metabolism	Loss of drug efficacy (if active inactive)
Vitamin B6 (pyridoxine)	Induce dopa decarboxylase ... metabolize levodopa in periphery	
Vitamin C	High doses of vitamin C (> 1000 mg/day) ... CYP enzyme induction Increase drug metabolism	
Charcoal-boiled meat (contains polycyclic aromatic hydrocarbons)	CYP Enzyme Inducer ... increase metabolism	
Food / nutrient (Enzyme inhibitors) Decrease drug metabolism		
Grapefruit	Most drugs ((CYP3A4 enzyme inhibitor decrease metabolism))	Increase drug toxicity (If activeinactive)
	Few drugs ((Albendazole, artemether, praziquantel, methyl prednisolone , Statins (except pravastatin), felodipine, nifedipine , verapamil, carbamazepine, diazepam, ethinynelestradiol, cyclosporine, sildenafil, ritonavir))	Increase drug Efficacy by inhibiting “ First-pass metabolism”
Lime juice	Same as grapefruit. If it is used to increase drug absorption, not less than 62 ml of the juice) should be administered with the drug	
Mango fruit	Contains high amount of Vitamin A which inhibit CYP2C19	Increase drug toxicity of anticoagulant (aspirin, warfarin, clopidogrel)
Watercress بقل , broccolis بروكولي,	Inhibit CYP2E1	Increase drug toxicity of acetaminophen, dapsone
Vitamin C	Deficiency in Vitamin C CYP enzyme inhibition.... Decrease drug metabolism	Increase drug toxicity

2.4. Influence of Food on drug Excretion

Drug excretion

It is the process by which the drug or its metabolites are removed from the body into body excretions through excretory organs as follows

Excretion from Excretory organs

- ✓ Urine kidney (Called renal excretion)
- ✓ Stool liver, intestine (also called hepatic excretion, biliary excretion)
- ✓ Sweat Sweat gland
- ✓ Saliva Salivary gland
- ✓ Exhale air lung
- ✓ Breast milk Breast

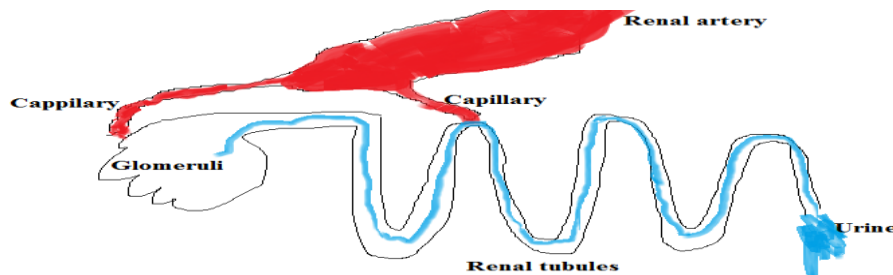
Steps of drug excretion

“ Drug transport from blood Excretion organ Excretion “

- ✓ Drug is distributed from blood to excretion organ
- ✓ Drug enter the organ to the excretion
- ✓ Drug is excreted with the excretion to out of the body

Mechanisms of renal drug excretion

- ✓ The functioning units in kidney are called “Nephrons”



The main mechanisms of drug excretion are

- 1- Glomerular filtration :** the drug transport from capillaries of the renal artery and enter “glomeruli” . This mechanism occurs for drugs (lipophilic, hydrophilic) with small molecular weight (< 500). It needs no energy or carrier
- 2- Active Tubular secretion:** the drug transport from capillaries of the renal artery and enter “proximal renal tubules” . This mechanism occurs for few drugs (hydrophilic only) with larger molecular weight (> 500). It needs energy or **carrier e.g.** MDR1 (Multidrug resistance 1) transporter Digoxin

Tubular Reabsorption : It is not a process of excretion. It is an opposite of excretion. The drug may get back (from urine back to blood) in the distal renal tubules. It occurs if

pH of urine	Drug)s that are reabsorbed (i.e. excretion decreased
More acidic	Acidic drugs e.g. .g. NSAIDS (e.g. Aspirin), cephalosporins, penicillins, barbiturates (e.g. phenobarbitone).
Alkaline	Basic drugs e.g. antihistamines (e.g. chlorpheniramine), antipsychotics (e.g. chlorpromazine)

Examples of food, nutrient affecting drug excretion

Food, nutrient	Mechanism	Clinical outcome
Citrus fruits (e.g. orange, lime) and vitamin C	make the pH of urine more acidic . Therefore, they can affect “Renal reabsorption” of some drugs	Increase tubular reabsorption (i.e. decrease renal excretion) of Acidic drugs Prolong duration and increase drug efficacy / toxicity
		Decrease tubular reabsorption (i.e. increase renal excretion) Decrease drug efficacy
Deficiency in vitamin C	Make the pH urine less acidic	Decrease “tubular reabsorption” increase excretion of acidic drugs e.g. barbiturates ...
Food that affect drug metabolism (See previous chapter)	Can also affect drug excretion	See previous chapter

CHAPTER III: INFLUENCE OF NUTRITIONAL STATUS ON DRUGS

3.1. Introduction

- ❖ **Normal** nutritional status of a person refers to a healthy status of weight (normal weight and also to normal level of nutrients (carbohydrates, proteins, lipids, minerals, vitamins and water) in the body.
- ❖ **Abnormal nutritional status** refers to the opposite of “normal status”. It includes:
 - Malnutrition
 - Obesity / overweight
 - Hypo or Hyper levels of minerals (e.g. hypocalcemia, hypercalcemia)
 - Hypovitaminosis or Hypervitaminosis
 - Hypo or hyper levels of macronutrients (e.g. Hypolipidemia, hypercholesteremia)
- ❖ **Body mass index (BMI)**

It is the index by which body weight and overall nutritional status is indicated as normal or abnormal

Nutritional status	Weight status	Body mass index (kg/m ²)
Abnormal : Below normal level (malnutrition)	Underweight	< 18.5
Normal	Normal	18.5 to < 25
Abnormal : Above normal level	Overweight	≥ 25 to < 30
	Obese	≥ 30

$$\text{BMI (kg/m}^2\text{)} = 10000 * (\text{weight kg}) / (\text{height cm})^2$$

$$\text{Or BM1 (kg/m}^2\text{)} = (\text{Weight kg}) / (\text{height cm} / 100)$$

BMI and interpretation of weight status in children

The determination of weight status based on BMI is different from that of adults. It depends on determination of the **BMI percentile** as follows:

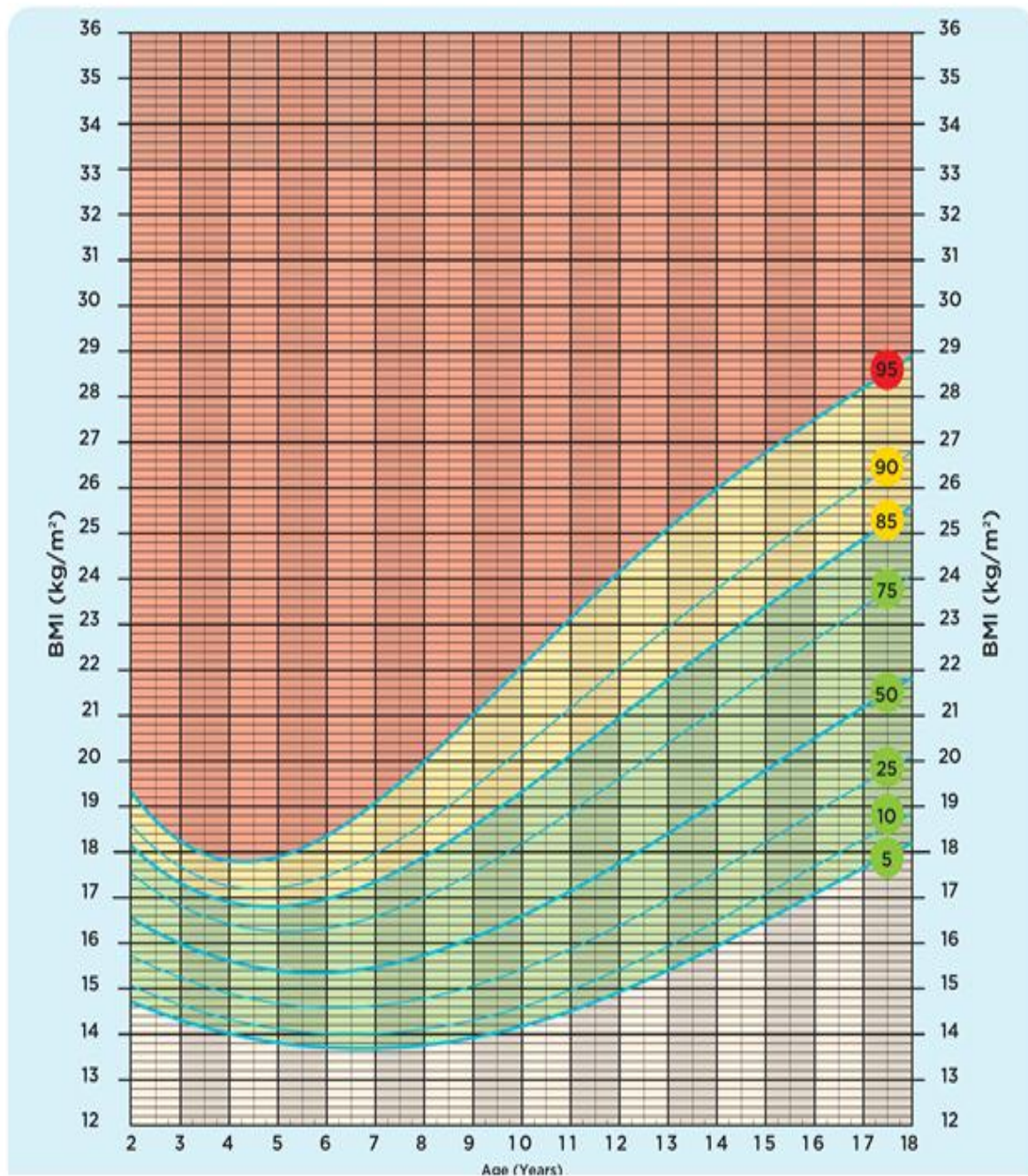
1. Calculate BMI using the same equation of adult

$$\text{BMI (kg/m}^2\text{)} = (\text{Weight kg}) / (\text{height cm} / 100)$$

2. Use the following graphical charts to determine BMI percentile (you can also use calculator at Medscape) for boys (2-18 years), or girls (2-18 years)

Body mass index (BMI)-for-age percentile chart

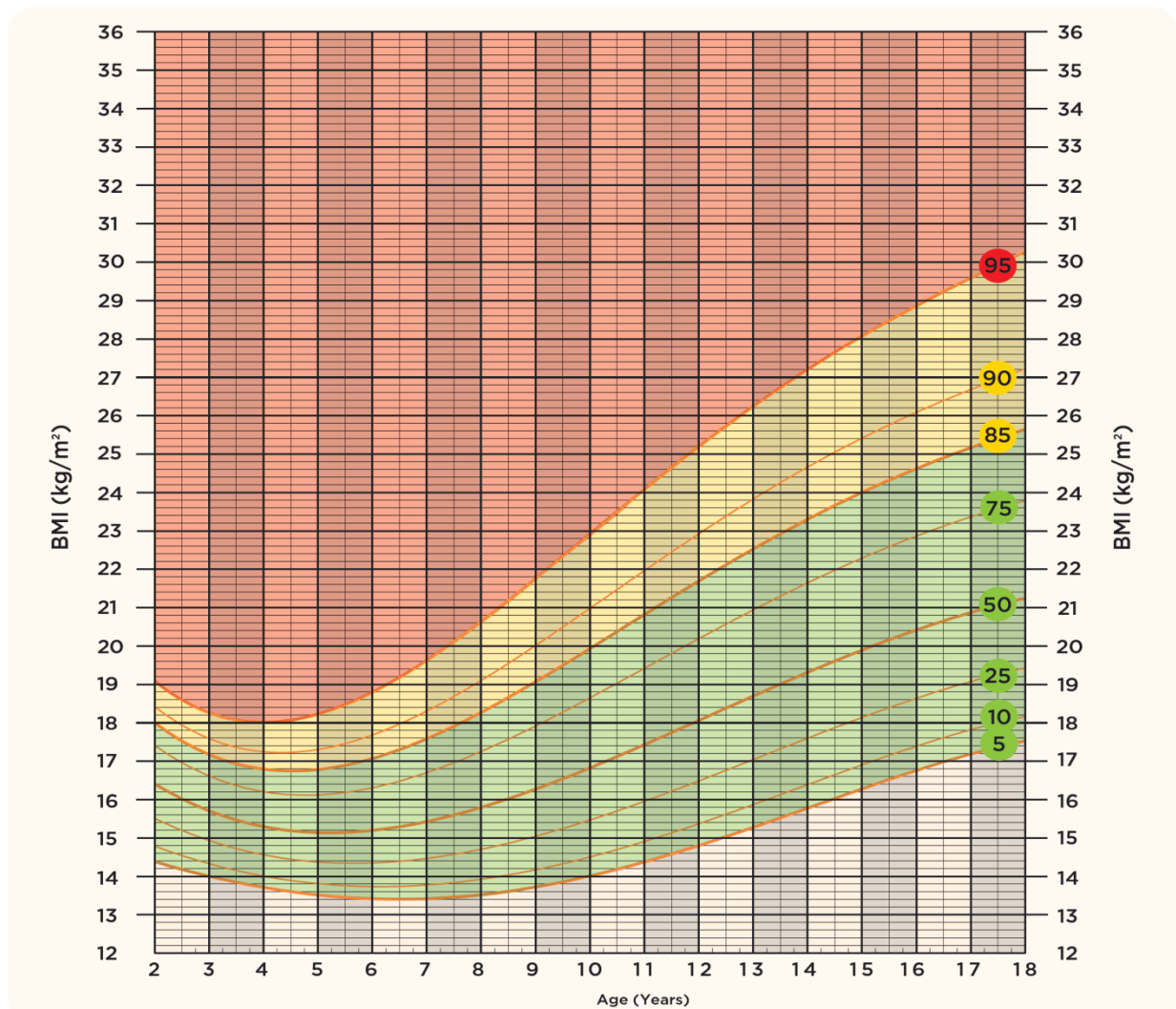
Boys)



For girls(2- 20 years)

Body mass index (BMI)-for-age percentile chart

Girls



3. Determine weight status using the following table

Body Mass Index Interpretation (boys or girls)

<i>Weight Status Category</i>	<i>BMI Percentile</i>
Underweight	<5th
Healthy weight	5th to <85th
Overweight	85th to <95th
Obese	≥95th

1.2. Influence of Malnutrition on drugs

❖ **Malnutrition Definition:** It is abnormal nutritional status exists if one or more nutrients are under the normal levels.

❖ **Causes of Malnutrition**

- **Primary :** Inadequate food intake
- **Secondary:** It occurs secondary to a disease examples:
 - acute illness in hospitalized patient.
 - GIT disorders : diarrhea, surgeries, sever inflammation and tumors of GIT
 - Loss of appetite and/or vomiting (e.g. in cancer patients receiving anticancers)
 - Sever Injuries or burns

Influence of malnutrition on drug disposition

Process	Mechanism	Clinical outcome
Drug absorption	✓ Malnutrition generally decreases drug absorption (i.e. cause malabsorption of drugs) especially in children. - Examples of drugs ✓ Chloroquine (antimalarial) ✓ Sulfadiazine (antitoxoplasmosis) ✓ Cyclosporine (Immunosuppressant) ✓ Antibiotics	Decrease drug efficacy
Drug distribution	Malnutrition generally decreases drug distribution how ? ✓ Reduces albumin synthesis decrease plasma protein binding of drug increase free drug increase drug elimination ✓ Decrease protein carriers (e.g. OATP) ... decrease distribution of drugs e.g. atorvastatin (hypolipidemic drug)	
Drug Metabolism	Malnutrition generally decreases drug metabolism how ? ○ Decreases the synthesis of hepatic enzymes (CYP) ○ Decreases hepatic blood flow Examples of affected drugs: ○ Analgesics (Aspirin , Acetaminophen) ○ General anesthetic (Ketamine) ○ Antiepileptic (phenytoin) ○ Antibiotics (Gentamicin , Clarithromycin) ○ Anti-TB (isoniazide) ○ Antimalarials (Quinine) ○ Antifungals (Itraconazole) ○ Anticancer (Doxorubicin , Methotrexate) - Diuretics (Furosemide and toresmide)	Increase drug toxicity
Drug Excretion	Unless the malnutrition causes decrease in renal function, there is no obvious relation between malnutrition and drug excretion.	Unchanged

3.2. Influence of obesity/overweight on drugs

1. Influence of obesity/overweight on drug disposition

Process	Mechanism	Clinical outcome of obesity on drug disposition
Drug absorption	limited data suggest that oral drug absorption, may be no different in obese individuals e.g., immunosuppressant (cyclosporine), sedative-hypnotic (midazolam), antibiotic (penicillin v) , antihypertensive <i>B</i> -blockers(propranolol)	Absorption unchanged
Drug distribution	Generally, influence of obesity on drug distribution is inconsistent and variable. Obesity causes increase in blood volume, cardiac output and organs masses, but cause no or limited change in blood flow and plasma protein binding of drugs. Therefore, the influence of obesity on drug distribution may be increase or decrease according to drug physicochemical properties. Obesity causes increase in drug distribution (i.e. greater V_D) for some Lipophilic drugs (e.g. digoxin , cyclosporine , phenytoin) . However, obesity may lower V_D for other lipophilic drugs (e.g. bisoprolol , diazepam) Obesity causes increase in drug distribution (greater V_D) for most Hydrophilic drugs (e.g. aminoglycosides , ampicillin , Cefotaxime , ciprofloxacin). However, obesity may lower V_D for some hydrophilic drugs (e.g. ranitidine , etrapenem)	Variable (increase / decrease)

Influence of obesity/overweight on drug disposition

Process	Mechanism	Clinical outcome of obesity on drug disposition
Drug Metabolism	Generally, influence of obesity on drug distribution is inconsistent and variable. Obesity may cause increase in metabolism of some drugs (e.g. prednisolone), decrease in metabolism of some drugs (e.g. methylprednisolone)	Variable (increase/decrease)
Drug Excretion	Renal drug clearance can be increased in obese patient for drugs that are partially or mainly excreted by tubular excretion(e.g., aminoglycosides, cefotaxime, ciprofloxacin, lithium, procainamide, piperacillin). However, drugs that are mainly excreted by glomerular filtration may have lower greater <u>or unchanged</u> excretion rate.	Variable (increase/ decrease)

Summary

Obesity, in general, causes no change in drug absorption, distribution, metabolism or excretion except for certain drugs as shown in the above table.

2. Influence of obesity/ overweight on drug efficacy

Obesity, in general, has variable influence on drug effect.

- ✓ Effect may increase : (e.g., glipizide, glyburide, prednisolone)
- ✓ Effect may decrease : (e.g. verapamil)

- ✓ **Drug dose in normal-weight, malnourished (underweight) , overweight/obese patients:**

- **In normal-weight patients**

The dose of a drug is usually based on body weight (**actual body weight** also called **total body weight TBW**).

- **In malnourished (underweight) patients**

Lean body mass (LBM) also known as Lean Body weight (LBW)

LBW is the weight of the body excluding the fat mass

For males $LBW (kg) = (0.73 * height\ cm) - 59.42$

For females $LBW (kg) = (0.65 * height\ cm) - 50.74$

Note : Using LBW, the drug dose is lower than normal dose used for normal-weight patient

- ✓ **In overweight/obese patients**

- ✓ In general, the dose of a drug in overweight depends on TBW similar to that of normal-weight patients.

- ✓ However, for some drugs whose drug distribution or drug clearance change significantly in overweight/obese, other types of weight formula may be necessary to calculate drug dose in overweight/obese patients.

Other type of weight formula can be used in overweight/obese patients

- ✓ **Lean body mass (LBM) also known as Lean Body weight (LBW)**

LBW is the weight of the body excluding the fat mass

For males $LBW (kg) = (0.73 * height\ cm) - 59.42$

For females $LBW (kg) = (0.65 * height\ cm) - 50.74$

Note : Using LBW, the drug dose is lower than normal dose used for normal-weight patient

✓ **Adjusted weight**

It is the total body weight multiplied by a correction factor

$$\text{(Adjusted weight (kg))} = \text{actual TBW kg} \times \text{Correction factor}$$

- ✓ Correction factor is difficult to calculate and usually depends on volume of distribution of the drug or its clearance in obese compared to normal-weight patient. There must be a reliable research to determine the correction factor.
- ✓ Correction factor may (< 1) or (> 1).

So, Using adjusted weight, the drug dose is lower (if correction factor < 1) or higher (if correction factor > 1) than normal dose used for normal-weight patient

Why ?

- ❖ Drugs whose distribution increases significantly in obese and can be stored in adipose e.g. phenytoin, needs a higher correction factor (1.5)
 - ❖ Drugs whose distribution increases significantly in obese but are hydrophilic and not stored in adipose e.g. gentamicin, needs a lower correction factor (0.45)
 - ❖ Drugs whose metabolism increases **but not significantly** e.g. prednisolone, needs no change in dose. i.e. use TBW
 - ❖ Drugs whose metabolism decreases significantly e.g. methylprednisolone, needs lower dose so we use LBW
- ✓ To determine which type of weight , you have to use in dose adjustment for or obese, search the literature to determine the change in drug clearance or drug distribution, Then use one of the following table. If information is unavailable use TBW

$Cl_{T\ OB}:Cl_{T\ Non-OB}$	Dosing Weight
≥ 1	Actual TBW
< 1	Adjusted or LBW

Or

$V_D/kg\ TBW_{OB}: V_D/kg\ TBW_{Non-OB}$	Dosing Weight
≥ 1	Actual TBW
0.7 up to 1	An adjusted body weight
< 0.7	LBW

Doses of some drugs in Obese patient

Drug	Weight on which dose is based	Dose in obese
Piperacillin	Actual TBW	Normal
Prednisolone	Actual TBW	Normal
Phenytoin	Adjusted (actual TBW x 1.5)	Increased
Aminoglycosides e.g. gentamicin	Adjusted (Actual TBW x 0.45)	Decreased
Methylprednisolone	LBW	Decreased

3.3. Influence of Vitamin and/or minerals on drugs

Bad interactions		
Vitamin / mineral	Mechanism	Clinical outcome
Vitamin D deficiency, Zinc deficiency	Decrease absorption of many drugs	Decrease or loss of drug efficacy
Vitamin B6 (pyridoxine)	Induce dopa decarboxylase ... metabolize levodopa in periphery	
Vitamin C	High doses of vitamin C (> 1000 mg/day) ... CYP enzyme induction Increase drug metabolism	
Vitamin K high intake	Antagonize drug efficacy of warfarin	
Niacin (nicotinamide B3), riboflavin (B2)	High doses.... CYP enzyme induction Increase drug metabolism ..	
Vitamin C	Deficiency in Vitamin C CYP enzyme inhibition.... Decrease drug metabolism	Increase drug toxicity
Vitamin A	High dose: increase the effect but may also increase toxicity of oral anticoagulants (aspirin, warfarin, clopidogrel) by inhibition CYP2C19	
Vitamin E	Increase antiplatelet toxicity of anticoagulant	
Positive interactions		
Vitamin / mineral	Mechanism	Clinical outcome
Vitamin B6	Replace vitamin B6 extensively metabolized by isoniazide ... decrease peripheral neuritis adverse effect	Decrease drug toxicity
Multivitamin (should include folic acid, vitamin B6, vitamin B12, vitamin B2, niacin (nicotinamide))	Decrease the drug toxicity (teratogenicity) of valproic acid in pregnant women	
Intake of vitamin D or Calcium	Decrease drug toxicity (risk of bone fractures) in patients taking antiepileptics (carbamazepine, valproic acid, phenytoin)	
Folic acid	Decrease drug toxicity of antifolate drugs (flurouracil, methotrexate , sulfonamides	

Positive interactions	Mechanism	Clinical outcome
Niacin (nicotinamide)	Increase the drug efficacy of "simvastatin"	Increase drug efficacy
Vitamin E	* Increase absorption of cyclosporine * Inhibit P-gp (efflux carrier) so increase the absorption of some drugs	
Calcium intake	Increase drug efficacy of antiepileptic (phenytoin, carbamazepine, phenobarbital etc.)	
Calcitriol (analogue of Vitamin D)	Increase efficacy of Docetaxel (anticancer)	

CHAPTER IV: INFLUENCE OF NONNUTRITIVE DIETARY ON DRUGS

Nonnutritive Dietary

Nonnutritive dietary include all what we eat, drink or inhale that don't contain any nutrient.

They are generally used to improve mood (euphoria), physical and mental activity.

Examples: **herbs, alcohol and tobacco smoke**

Nonnutritive dietary	Mechanism	Clinical outcome
Herbs (Increase Drug toxicity)		
Black tea	Inhibit CYP2E1	Increase Drug toxicity of (acetaminophen, dapsone)
Tea, coffee , chocolate Cola.	All these herbs contain methylxanthines e.g. (theophylline, caffeine) ... CYP Enzyme inhibitors ... decrease metabolism	Increase drug toxicity (If activeinactive)
Garlic Used as flavoring ingredient in food. Also used to treat hypertension, atherosclerosis, hypercholesterolemia	Inhibit thromboxane A2 Inhibit platelet aggregation	High intake Increase toxicity of anticoagulants (aspirin, clopidogrel, ticlopidine)
	CYP2E1 inhibitor	Increase drug toxicity of acetaminophen, ethanol, inhalation general anesthetics (e.g. desflurane)
Ginger	Unknown, but increase drug efficacy	* Increase drug toxicity (bleeding) of oral anticoagulants. * Increase drug toxicity (hypoglycemia) of antidiabetics
Licorice	Increase sodium retention, hypertension, hypokalemia	Digoxin: increase toxicity (severe hypokalemia * Diuretic hydrochlorothiazide : increase drug toxicity (severe hypokalemia)
Ginkgo	Inhibit PAF (platelet aggregating factor)	Increase toxicity of oral anticoagulant (Bleeding)

Dong quai (<i>Angelica sinensis</i>), danshen (<i>Salvia miltiorrhiza</i>),	CYP Enzyme inhibitor ... decrease metabolism	Increase Toxicity of warfarin (bleeding)
Kava (<i>Piper methysticum</i>) A type of Pepper فلفل Also used as anxiolytic	CYP inhibitor	Increase drug toxicity of alprazolam (Coma), levodopa (mental deterioration) and other drugs. Kava extract also cause hepatotoxicity
Valerian الناردين (<i>Valeriana officinalis</i>)	Stimulate GABA receptors	Induce toxic actions of GABA agonists (barbiturates, benzodiazepines)
Herbs (Decreasing drug efficacy)		
Khat	Can decrease absorption Of many drugs	Cause decrease absorption of many drugs e.g. ciprofloxacin
Tea, coffe, chocolate, Cola	Contain Methylxanthines have diuretic effect	Decrease blood level of minerals(Calcium, magnesium, sodium) ... by increasing their excretion
Ginseng	Induces CYP2C9	Decrease drug efficacy of warfarin
	It may contain germanium (toxic heavy metal) which cause renal insufficiency	Decrease diuretic effect of furosemide and may cause renal failure
St. John's wort (SJW) (<i>Hypericum perforatum</i>) Used to treat wound and burns and mild depression	CYP3A4 Enzyme inducer	It increases first-pass metabolism also increase general metabolism of many drugs
Herbs (Increasing drug efficacy)		
Nuts, seeds	Contain plant sterols and stanols	Increase drug efficacy of "Statins" hypolipidemic drugs
Garlic	CYP2E1 inhibitor	*Inhibit first-pass metabolism of some drugs e.g. saquinavir So increase drug efficacy
	Inhibit thromboxane A2 Inhibit platelet aggregation	Mild-moderate intake Increase efficacy of anticoagulants (aspirin, clopidogrel, ticlopidine)

Non-herbal (Increase drug toxicity)		
Glucosamine/chondroitin “ dietary supplement used for osteoarthritis and joint disorders”	Unknown	Increase drug toxicity of warfarin (bleeding)
Alcohol	Some drugs [e.g. metronidazole, sulfonyleureas, some cephalosporin antibiotics, and ketoconazole] can inhibit (aldehyde dehydrogenase) which is necessary to metabolize alcohol	Interaction causes severe adverse effect (disulfiram)
	Acute intake : CYP2E1 inhibition	Increase toxicity of some drugs e.g. acetaminophen ethanol, inhalation general anesthetics (e.g. halothane)
Non-herbal (Decrease drug efficacy)		
tobacco smoking (contains polycyclic aromatic hydrocarbon)	CYP3A4 Enzyme inducer	Loss of drug efficacy (if active inactive)
Alcohol	Chronic small intake : CYP2E1 induction	Decrease drug efficacy of some drugs, inhalation general anesthetics (e.g. halothane)

CHAPTER V: FOOD-DRUG INTERACTIONS IN SPECIFIC NUTRITION

5.1. Introduction

- ❖ **In general**, oral (swallowing) nutrition is the best nutrition for patients.
- ❖ **Non-oral nutrition**
 - Enteral nutrition (tube feeding)
 - Total parenteral nutrition (TPN) are necessary.
- ❖ **Reason for non-oral nutrition**
 - Oral nutrition is inappropriate for the patients
 - Acute malnutrition
 - Patients who have been without oral intake for long time :
 - ✓ Children (3- 5 days)
 - ✓ Adult (5- 10 days)

5.2. Enteral nutrition (EN) (Tube feeding)

- ❖ **Definition:** Food is given in liquid form through specific tube (nasal .. pharynx ... stomach).
- ❖ **Contraindications of EN**
 - Peritonitis
 - Sever and intractable vomiting
 - Sever and Intractable diarrhea
 - Ischemia of intestine

❖ Guidelines for drug administration in patients on EN

1- Patients on EN are better receives **parenteral** (injectable) drugs.

2- However, some drugs are unavailable as injections so what should we do ?

(i) **(Oral :** Administer drugs orally whenever possible

(ii) **Other routes :** Administer the drug by other routes (rectal, sublingual , buccal, transdermal) if the drug dosage form is available by these routes

(iii) **Through tube :**

- **Don't mix** the drug with EN (in most cases especially if food decreases drug absorption). You may mix the drug with EN If the drug if food increase drug absorption or if food decrease GIT upset. See previous chapters
- **Liquid dosage forms (e.g. solution, suspension, emulsion):** If the drug that are available as liquid dosage form, it can be given by tube feeding by dilution the dose with 30-50 ml water.
- **Solid dosage forms (e.g. tablets, capsules) .** If the drug is only available as solid dosage form :
 - ✓ **Powder for suspension (e.g. pediatric antibiotic):** Prepare the suspension by adding water to a definite mark. Shake and take the recommended dose. Dilute the dose with 30-50 ml water.
 - ✓ **Effervescent granules/powders” :** dissolve in 30-50 ml water
 - ✓ **Tablet :** Grind the tablet then mix it with warm water (30-50 ml).
 - ✓ **Capsule: Hard gelatin capsules :** empty the capsule powder then mix it with warm water (30-50 ml). If the capsule contains granules, it must be grinded first then mix it with (30-50 ml) warm water. **Soft gelatin capsules:** Dissolve the whole capsule in 30-50 ml warm water
 - ✓ **Solid dosage forms that can not be added into EN**

- **Tablets** : Extended-release tablets or enteric coated tablets or sugar-coated tablets.
- **Capsules**: that contain sustained or extended formulations (e.g. beads)
- **When two drugs are to be administered**
 - Don't mix the two drugs together
 - If the two drugs are to be given at the same time :
 - * administer the first drug (as described before)
 - * Give a flush of 15 -30 ml water into the tube
 - * administer the second drug (as described before)
- **Type of water used to flush tube or to prepare drugs**
 - **Drinking water** (the best)
 - **Sterile water**: for patients having severe infections or in immunocompromised patients (e.g. AIDS patients, patients using immunosuppressant drugs for autoimmune diseases or organ transplantation)
- **Before and After finishing drug administration**
 - **Flush the tube** with 15-30 ml water
 - **Hold feeding** : For Drugs that should not be added to EN (Drugs that should NOT be given with food), hold feeding (don't give EN) for at least 1 hour before and 2 hours after drug administration.

Food in EN influence on with drugs

- 1- **Food, nutrients, non-nutritive dietary** (same as mentioned in the previous chapters) can interact with drugs if food is given by EN
- 2- **Proteins** Whey protein is unlikely to cause interactions with drugs than other types of proteins
- 3- **Oils** : Especially high-fat can cause major interactions with drugs
- 4- **The feeding tube** usually contain (Polyvinyl alcohol) which can adhere to phenytoin suspension ... Decreasing drug absorption Loss of drug efficacy

5.3. Parenteral nutrition (PN) (Total parenteral nutrition; TPN)**Definition**

It is administration of nutrients totally by intravenous route.

Components of TPN

- 1- **Macronutrients** : carbohydrates (dextrose), proteins (amino acids), Fat
- 2- **Micronutrients**:
 - **Vitamins** (fat soluble: A, D, E, and K.), (Water soluble: thiamin B1, riboflavin B2, niacin (nicotinamide), pyridoxine B6, vitamin B12, folic acid, biotin, pantothenic acid, and ascorbic acid C)
 - **Minerals**
 - **Essential Electrolytes** (Sodium, potassium, magnesium, calcium) salts
 - **Trace elements**: zinc, copper, chromium, manganese, and selenium)
- 3- **Water**

Indications of TPN

When oral or enteral nutrition are inappropriate or contraindicated. Examples:

- Complete intestinal obstruction
- Paralytic ileus (unless localized to the stomach)
- High-output intestinal fistula (volume >500 mL/day)
- Severe, intractable diarrhea (volume >1500 mL/day)
- Short bowel syndrome
- Severe acute pancreatitis
- Malnutrition (unable to take oral or enteral nutrition)
- Coma

Types of TPN formulation

1- 2-in-1 formulation

It is the combination of the two macronutrients dextrose and amino acids along with electrolytes, vitamins, and trace elements in the same infusion container while the intravenous fat emulsions (IVFE) are administered separately.

Composition of intravenous fat emulsion drip (1 Liter) in 2-1 formulation

<i>Component or Characteristic</i>	<i>Intralipid (Baxter)</i>	<i>Liposyn III (Hospira)</i>
Oil (%)	Soybean (10, 20, 30)	Soybean (10, 20, 30)
Egg yolk phosphatide (%)	1.2	1.2
Glycerin (%)	2.25	2.5
Fatty acids (%)		
Linoleic acid	50	54.5
Oleic acid	26	22.4
Palmitic acid	10	10.5
Linolenic acid	5	8.3
Stearic acid	3.5	4.2
Osmolarity (mOsm/L)	260–268	284–292
Approximate pH	8	8.3
Fat particle size (µm)	0.5	0.4

Intralipid , Liposyn are (brand names). Baxter and Hospira are (Pharmaceutical companies)

Composition of other nutrients intravenous drip (1 Liter) in 2-in-1 formulation

<i>Component and Unit</i>	<i>Formulas</i>	
	<i>Peripheral Line</i>	<i>Central Line</i>
Amino acids Injection Product (g)	35	42.5
Dextrose (g)	50	250
Multivitamins (mL)	10	10
Trace elements (mL)	1	1
Electrolyte content		
Sodium (mEq)	41	44
Potassium (mEq)	40	30–40
Calcium (mEq)	5–9	5–9
Magnesium (mEq)	8	8
Chloride (mEq)	65–78	43–65
Acetate (mEq)	25–31	30–38
Phosphorous (mmol)	3.5	10–15

Peripheral : administered into a peripheral vein, usually of the hand or forearm.

Central : into a large diameter vein, usually the superior vena cava adjacent to the right atrium.

2- 3-in- 1 formulations

All three macronutrients in one infusion container) . This formula is also known as “total nutrient admixture” (TNA)

Composition of nutrients intravenous drip (1 Liter) in 3-in-1 formulation

<i>Component and Unit</i>	<i>Formulas</i>	
	<i>Peripheral Line</i>	<i>Central Line</i>
Amino acids injection product (g)	30	49
Dextrose (g)	50	200
IVFE (g)	20	35
Multivitamins (mL)	10	10
Trace elements (mL)	1	1
Electrolyte content		
Sodium (mEq)	43	40
Potassium (mEq)	40	40
Calcium (mEq)	5-9	5-9
Magnesium (mEq)	8	8
Chloride (mEq)	35-45	35-45
Acetate (mEq)	42-52	45-68
Phosphorous (mmol)	7.5-15	10-15

Peripheral : administered into a peripheral vein, usually of the hand or forearm.

Central : into a large diameter vein, usually the superior vena cava adjacent to the right atrium.

Total infused fluid (TIF)

- ❖ **TIF:** The total volume of 2-in-1 or 3-in-1 formula of TPN to be administered daily
- ❖ **Normal nutritional status patients :** Patients (50- 100 kg body weight), administer daily 1.6- 2.6 L According to the following equation

$$\text{TIF} = [(\text{Patient weight kg} - 20) \times 20] + 1500$$

- ❖ **Patients with specific abnormal nutritional status :** May need adjustment of each nutrient.

Food-drug interaction in TPN

1. Physical Incompatibility of drugs mixed into TPN

It could cause physical interactions (Precipitation, gas evolution, color change) which can seriously affect drug efficacy or increase its toxicity

Y-Site Injection Compatibility (1:1 Mixture): Drugs Incompatible with 2-in-1 Solutions

Acyclovir	Doxorubicin	Metoclopramide HCl	Potassium phosphate
Amphotericin B	Fluorouracil	Midazolam HCl	Promethazine HCl
Cefazolin sodium	Furosemide	Minocycline HCl	Sodium bicarbonate
Cisplatin	Ganciclovir sodium	Mitoxantrone HCl	Sodium phosphate
Cyclosporine	Immunoglobulin	Phenytoin sodium	
Cytarabine	Methotrexate sodium		

Y-Site Injection Compatibility (1:1 Mixture): Drugs Incompatible with 3-in-1 (TNA) Formulations

Acyclovir	Ganciclovir	Lorazepam	Pentobarbital
Amphotericin B	sodium	Methyldopate HCl	sodium
Cyclosporine	Haloperidol	Midazolam HCl	Phenobarbital
Dopamine HCl	Heparin	Minocycline HCl	sodium
Doxorubicin	Hydrochloric acid	Morphine ^a sulfate	Phenytoin sodium
Doxycycline hyclate	Hydromorphone	Nalbuphine HCl	Potassium phosphate
Droperidol	Iron dextran	Ondansetron HCl	Sodium phosphate
Fluorouracil	Levorphanol tartrate		

2. Pharmacological Incompatibility of drugs

TPN contains nutrients and therefore they may affect drug disposition and efficacy See page (19, 23, 30, 31) . Example: TPN contains vitamins (E) that can increase toxicity of anticoagulants.

3- Long-term use of TPN

- ❖ Generally decrease liver function (Decrease CYP synthesis)
Decrease Drug metabolism.....Increase drug toxicity (If active
inactive
- ❖ Decrease CYP activity Decrease drug metabolism (by increasing
level of pro-inflammatory (cytokines) Increase drug toxicity (If
active inactive
- ❖ Increases the volume of total body and extracellular fluid. This can
increase drugs distribution to non-target tissues (e.g., aminoglycoside
and b-lactam antibiotics) Increase side effect of the drug (Increase
drug toxicity)

CHAPTER VI: DRUGS INTERACTING WITH NUTRIENTS

6.1. Introduction

As described in the previous chapters, food , nutrients or nonnutritive dietary can interact with drugs affecting their efficacy or toxicity. However, in this chapter, the influence of drugs on nutrients is discussed .

Many drugs cause **side effects** as change in body weight (weight gain , weight loss) or change the levels of nutrients in the body (hypocalcaemia, hypovitaminosis, etc.).

Intervention of the therapeutic nutritionist

For drugs that cause side effects that change the nutritional status of a patient, the intervention of a therapeutic nutrition specialist is important help the patient to manage those side effects

- ✓ **No drug stop/No drug dose change** : Never advice the patient to stop/change drug **but only under** physician permission. As a nutritionist, you may provide a “**suggestion**” to the physician for alternative drugs.
- ✓ **Diet** : Give the patient a list of diet that should include “**Food /dietary supplement to be taken**” and “**Food/dietary supplement to be avoided**”
- ✓ **Exercise:** advice the patient to perform/avoid certain exercises that can help to alleviate the side effect

6.2. Drugs causing Weight gain (increase body weight)

- ✓ Some drugs can increase the body weight causing “Weight gain”
- ✓ If the weight gain is Major and progressive, intervention of the therapeutic nutritionist

Mechanism of Drug-inducing weight gain (How drugs can cause weight gain ?)

- ✓ **Stimulate appetite** (Appetizer drugs) e.g. cyproheptadine. They increase the filling of hunger. Some drugs may improve the psychological status/mood which e.g. antipsychotics (e.g. chlorpromazine), antidepressants (e.g. amitriptyline)
- ✓ **Decrease the catabolism** of macronutrients (e.g. inhibit lipolysis, inhibit protein degradation in muscles)
- ✓ **Increase absorption** of macronutrients from food.
- ✓ **Increase “biochemical shift”** of carbohydrate to fats
- ✓ **Increase distribution/storage** of macronutrients to a adipose tissue, muscles and liver

Drugs causing weight gain		Mechanism and Clinical outcome
Appetizers	Cyproheptadine	Increase appetite (the desire to eat)
Antipsychotics	chlorpromazine, clozapine, olanzapine	Patients experiencing relief from symptoms of depression and psychosis may eat more or overindulge.
Antidepressants	amitriptyline, imipramine, mirtazapine, risperidone, and ziprasidone	
Antidiabetic drugs	Insulin , sulfonylureas (e.g. Glibenclamide, Glipizide, etc.), and thiazolidinediones (e.g. pioglitazone)	✓ Inhibit lipolysis in adipose tissue ✓ Stimulate amino acid and glucose transport into muscles . ✓ Increase shift of carbohydrate to fats. ✓ Studies have reported weight increases from 0.8 to 6.6 kg after the start of the medication. Most weight gain reached a plateau (maximum) effect by 6–12 months

Drugs causing weight gain			Mechanism & clinical outcome
Steroids	Male hormones	Testosterone, methyltestosterone	✓ Inhibit lipolysis in adipose tissue, stimulate amino acid and glucose transport into muscles, Increase protein synthesis, edema ✓ Oxandrolone is approved by FDA to promote weight gain in patients who have lost weight as a result of chronic infection, surgery, or severe trauma
	Anabolic steroids	anabolic steroids e.g. Oxandrolone	
	Female hormones	Oral contraceptives (estrogen) , Progestin-derivative (Megestrol)	Increase protein synthesis, edema
	Corticosteroids	E.g. dexamethasone , prednisolone	✓ Increase appetite ✓ Cause fluid retention ✓ Hyperlipidemia ✓ Increase storage of fat in abdomen, face and neck.
Antiepileptics		Valproic , carbamazepine, gabapentin	Various mechanisms
B-blockers		Propranolol, metoprolol	

Advices to patients suffering from drugs-induced weight gain

1- Diet

- Snacks : 6 small meals / day to suppress hunger
- Proteins: 25-30 gram/ meal
- Unprocessed food that should include: Fruits, vegetables, yoghurt, avocado, nuts and plant oil
- Avoid trans fats, high-calorie carbohydrates, avoid package/processed food,

2- Exercises: 150 min/week of moderate exercise

6.3. Drugs causing Weight loss (Decrease body weight)

Mechanism of Drug-inducing weigh Los (How drugs can cause weight loss ?)

- ✓ Inhibit appetite (Anorexic drugs).
- ✓ Alter taste perception
- ✓ Causing dry mouth
- ✓ Cause dysphagia
- ✓ Cause change in GIT functions

6.3.1. Anorexic drugs

1- Drugs that inhibit appetite (Anorexic drugs) as side effect	
✓ Ergotamine (anti-Migraine) ✓ Caffeine : used as adjuvant to acetaminophen and ergotamine to potentiate analgesic effect. ✓ Theophylline : (used for bronchial asthma). ✓ Metformin ✓ Opioid analgesic e.g. tramadol ✓ Antiepileptic Toprimate, lamotrigine ✓ Serotonergic drugs (Serotonin agonists) e.g. & antidepressants (Selective Serotonin reuptake Inhibitors ; SSRI) e.g. Fluoxetine , citalopram, fluvoxamine, paroxetine, sertraline ✓ Metronidazole ✓ Alcohol, smok	Most of these drugs have CNS stimulation properties .
2- Drugs approved by FDA to inhibit appetite (Anorexic drugs).	
✓ Liraglutide ✓ Phentermine -Topiramate ✓ Naltrexone-Bupropion	Used in obesity .

6.3.2. Drugs causing altered taste perception

Drugs that induce changes to an individual's perception of taste can result in decreased oral intake and weight loss. The effect of these drugs can not only due to direct contact with tongue, but also it occurs after drug distribution to chemosensory nerves of tongue causing disruption of taste sense.

Types of altered taste perception:

- **Dysgeusia** : taste of most food become : metallic, sour, bitter , salty or sweet.
- **Hypogeusia**: One or more taste perception decrease.
- **Ageusia** : loss of taste perception
- **Phantogeusia** : Decrease ability to recognize specific tastes

Examples of drugs that cause dysgeusia	
Type of dysguesia	Examples of drugs
Sour (acidic)	Losartan, , Acetazolamide
Metallic	Metronidazole , allopurinol, ethambutol, , enalapril , iodine, imethyldopa , sulindac , tetracycline
Salty	Losartan, Amiloride
Bitter	Captopril, aspirin , levodopa
Sweet	Losartan, Sulfasalazine , Furosemide

Examples of drugs that cause ageusia
✓ Amiloride ✓ Baclofen ✓ Carbamazepine ✓ Captopril ✓ Diazoxide ✓ Dilitiazem, ✓ Dicyclomine ✓ Enalapril ✓ Hyoscine ✓ Insulin ✓ Lomefloxacin

Management of altered taste perception

- **Masking techniques** including chewing sugarless gum or using lozenges or breath mints to help alleviate dry mouth or altered taste may be tried.
- **Zinc supplement** : used only if “Zinc deficiency” is the result of altered taste perception

6.3.3. Drugs causing Dry mouth

These drugs can cause dry mouth and thereby increasing water oral intake by the patient. This increases fullness of stomach and can suppress desire to eat

Examples of drugs that cause dry mouth
✓ Anticholinergic drugs ✓ Examples: Hyoscine, Dicyclomine, Atropine ✓ Drugs that have anticholinergic as additional action ○ Antidepressants e.g. imipramine, amitriptyline ○ Antihistamines e.g. meclizine

6.3.4. Drugs causing Dysphagia (Difficulty in swallowing)

These drugs can cause difficulty/pain during swallowing which can suppresses the patient's desire to eat

Examples of drugs that cause dry mouth
○ Most anticancers ○ Doxycycline ○ Iron supplement ○ NSAIDs ○ Potassium supplement

6.3.5. Drugs that change GIT functions

Drugs that cause physiological changes in GIT can result in decrease/increase food intake and thereby change nutritional status of the patients

6.3.5.1. Drugs induce emesis (vomiting)

These drugs can cause nausea and vomiting leading to decreasing the appetite and decreasing the absorption of nutrients. The most common emetogenic drugs are the Cytotoxic chemotherapy anticancers e.g. (**carmustine, cisplatin, cyclophosphamide, daunorubicin, doxorubicin, epirubicin, idarubicin, ifosfamide, irinotecan, lomustine**). Other drugs are: dopamine agonists (e.g. bromocriptine), Digoxin, metformin, SSRIs and sulfasalazine.

6.3.5.2. Drugs increasing GIT motility

These drugs can increase GIT motility leading to

- Acceleration gastric emptying time. This can results in decreasing nutrient release, dissolution and absorption
- Increase intestinal muscle contraction: Causing abdominal colic or diarrhea. They can greatly decrease nutrient absorption

Examples of drugs increasing GIT motility
○ Most NSAIDs: cause also gastric irritation ○ Anti-emetics: metoclopramide, domperidone ○ Laxatives e.g. bisacodyl, castor oil ○ Antibiotics (Broad spectrum)

6.3.5.3. Drugs Decreasing GIT motility

These drugs can decrease GIT motility leading to

- Acceleration gastric emptying time. This can results in delay nutrient absorption
- Decrease intestinal muscle peristalsis: causing constipation also delay nutrient absorption

Examples of drugs increasing GIT motility

- **Anticholinergics:** e.g. atropine, hyoscine,
- **Opiates and opoids:** e.g. morphine, loperamide, diphenoxylate
- **Anti-depressants:** e.g. amitriptyline, imipramine

6.4. Drugs decreasing the levels of nutrients in the body

6.4.1. Drugs causing Hypoglycemia

They decrease the blood glucose level below normal value.

Examples of drugs causing hypoglycemia

- **Antidiabetic drugs** (Insulin, oral hypoglycemics)
- **Alcohol (Excessive intake)**
- **Few Antidepressants (Some SSRIs ; venlafaxine)**
- **Androgens: (testosterone), anabolic steroids (oxandrolone), danazol**
- **Angiotensin-converting enzyme inhibitors (e.g. Captopril)**
- **Calcium channel blockers e.g. amlodipine**
- **Salicylates Aspirin)**
- **Antibiotic (tetracycline)**
- **Oral anticoagulant (warfarin)**

6.4.2. Drugs causing hypolipidemia

These are drugs used to treat hyperlipidemia . They are called “**Hypolipidemic drugs**”.

They decrease the blood lipid (total cholesterol, LDL, triglycerides) but increase HDL

At standard doses and diet-control, these drugs are not likely to cause hypolipidemia . However, extensive use of these drugs with no diet-control may cause hypoilipidemia.

Examples of drugs causing Hypolipidemia

- **Statins:** atorvastatin, pravastatin, lovastatin, simvastatin , rousvastatin
- Fenofibrate
- Gemfibrozil
- Cholestyramine
- Niacin (nicotinamide) (vitamin B3)

6.4.3. Drugs causing protein depletion (Hypoproteinemia)

They decrease the normal level of protein in the body..

Examples of drugs causing protein loss

- **Corticosteroids** : decrease the synthesis of growth hormone (Depletion of protein)
- **Alcohol** : inhibits protein absorption, increase nitrogen excretion

6.4.4. Drugs causing Hyponatremia

They decrease the normal level of sodium in the blood.

Examples of drugs causing hyponatremia

- Diuretics
- Drugs causing emesis (vomiting) e.g. **carmustine, cisplatin, cyclophosphamide, daunorubicin, doxorubicin, epirubicin, idarubicin, ifosfamide, irinotecan, lomustine**

6.4.5. Drugs causing Hypocalcaemia

They decrease the normal level of calcium in the blood.

Examples of drugs causing hypocalcaemia

- Anti-TB (Rifampicin)
- Antiepileptics (phenytoin, phenobarbitone)
- Bisphosphonate (e.g. ibandronate)
- Calcitonin
- Estrogens
- Corticosteroids
- Diuretics (loop diuretic e.g. furosemide, torsemide)
- Proton-pump inhibitors (e.g. omeprazole, lansoprazole)
- Digoxin
- Tetracycline
- laxatives

6.4.6. Drugs causing Hypomagnesemia

They decrease the normal level of magnesium in the blood.

Examples of drugs causing hypomagnesemia

- Proton-pump inhibitors
- Diuretics
- Digoxin
- Corticosteroids
- Oral contraceptives
- Tetracyclines
- laxatives

6.4.7. Drugs causing Hypophosphatemia

They decrease the normal level of phosphate in the blood.

Examples of drugs causing hypophosphatemia

- Antacids
- Excessive calcium supplement
- Excessive magnesium intake
- Laxatives

6.4.8. Drugs causing hypokalemia

They decrease the normal level of potassium in the blood.

Examples of drugs causing hypokalemia	
○	Diuretics
○	Insulin (High dose) increase uptake of potassium to cardiac muscles (cardiac arrhythmia, cardiac arrest)
○	Corticosteroids
○	Digoxin
○	B2-agonists e.g. salbutamol, ephedrine
○	Methylxanthines e.g. caffeine, theophylline
○	Calcium channel blockers verapamil
○	laxatives

6.4.9. Drugs causing deficiency of trace elements

Zinc Deficiency

Antivirals	Estrogen	Potassium-sparing diuretics
Captopril	Ethambutol	Thiazides
Corticosteroids	Folic acid	Valproic acid
Deferiprone	Hydrochlorothiazide	
Edetate calcium disodium	Oral contraceptives	

Iron Deficiency

Aspirin	Erythropoietins	Sulfonamides
Calcium	Ethanol	Tetracyclines
Cholestyramine	Indomethacin	
Deferoxamine	Neomycin	

Copper Deficiency

Antacids	Ethambutol
Antivirals	Zinc salts

6.4.10. Drugs causing Hypovitaminosis (Vitamins deficiency)

Drug-Induced Vitamin Depletion

Vitamin Deficiencies

Folic Acid Deficiency

Aspirin	Indomethacin	Primidone
Carbamazepine	Methotrexate	Pyridoxine
Celecoxib	NSAIDs	Sulfasalazine
Cholestyramine	Oral contraceptives	Valproic acid
Corticosteroids	Phenytoin	
H2 blockers	Potassium-sparing diuretics	

Vitamin A (Retinol) Deficiency

Cholestyramine	Mineral oil
Ethanol	Neomycin
	Orlistat

Vitamin B₁ (Thiamin) Deficiency

Aminoglycosides	Ethanol	Phenytoin
Cephalosporins	Fluoroquinolones	Sulfonamides
Digoxin	Loop diuretics	Tetracyclines

Vitamin B₂ (Riboflavin) Deficiency

Aminoglycosides	Oral contraceptives	Tetracyclines
Cephalosporins	Phenothiazines	
Fluoroquinolones	Sulfonamides	

Vitamin B₃ (Niacin) Deficiency

Aminoglycosides	Ethanol	Sulfonamides
Cephalosporins	Fluoroquinolones	Tetracyclines
Cholestyramine	Isoniazid	Valproic acid
Colestipol		

Vitamin B₆ (Pyridoxine) Deficiency

Aminoglycosides	Fluoroquinolones	Oral contraceptives
Cephalosporins	Hydralazine	Sulfonamides
Estrogen	Isoniazid	Tetracyclines
Ethanol	Loop diuretics	Theophylline

Vitamin B₁₂ (Cyanocobalamin) Deficiency

Aminoglycosides	Ethanol	Oral contraceptives
Antivirals	Fluoroquinolones	Phenytoin
Cephalosporins	H2 blockers	Proton pump inhibitors
Cholestyramine	Metformin	Sulfonamides
Colchicine	Neomycin	Tetracyclines

*Vitamin Deficiencies***Vitamin C Deficiency**

Aspirin

Vitamin D Deficiency

Mineral oil

Orlistat

Vitamin E Deficiency

Aspirin

Cholestyramine

Colestipol

Corticosteroids

Loop diuretics

Oral contraceptives

Orlistat

Vitamin K Deficiency

Aminoglycosides

Barbiturates

Cephalosporins (cefotetan)

Cefoperazone

Cholestyramine

Fluoroquinolones

Mineral oil

Orlistat

Phenytoin

Sulfonamides

Tetracyclines

6.5. Drugs causing increase nutrients level in body**6.5.1. Drugs causing Hyperglycemia**

- ✓ They increase the blood glucose level above normal value.
- ✓ They increase the risk of diabetes

Examples of drugs causing hyperglycemia

- Antipsychotic drugs (The atypical or second-generation) including clozapine, olanzapine, and risperidone, have a strong diabetogenic
- B-blockers : e.g. propranolol can inhibit insulin release causing severe hyperglycemia in Diabetes Type II
- Corticosteroids: decrease peripheral uptake of glucose by body tissues.
- Oral contraceptives: decrease sensitivity of insulin receptors
- Immunosuppressant (e.g. Cyclosporine)
- Sympathomimetic amines (e.g. ephedrine)
- thiazide diuretics (e.g. hydrochlorothiazide)
- Methylxanthines (e.g. theophylline)
- Thyroid hormone (Levothyroxine)

6.5.2. Drugs causing Hyperlipidemia (Hypercholesterolemia)

- ✓ They Increase the blood lipid (total cholesterol, LDL, HDL, triglycerides) level above normal value. Some drugs increase LDL, triglycerides but decrease HDL e.g. Protease Inhibitors (Anti-AIDS) e.g. Saquinavir..

Examples of drugs causing Hyperlipidemia

(They cause increase of lipid in blood by 40% or more than normal levels)

- Anabolic Agents (Growth hormone, anabolic steroids, Insulin)
- Corticosteroids
- CVS drugs: Antihypertensives : Beta-Blockers (e.g. propranolol) Diuretics (e.g. hydrochlorothiazide), antiarrhythmics (e.g. amiodarone),
- Protease Inhibitors (Anti-AIDS) e.g. Saquinavir, ritonavir
- Immunosuppressive Agents (e.g. cyclosporine, corticosteroids)
- Anticonvulsants (e.g. carbamazepine)
- Isotretinoin (analogue of Vitamin A) used orally

6.5.3. Drugs causing Hypernatremia

They increase the normal level of sodium in the blood.

Examples of drugs causing Hypernatremia

- Laxatives

6.5.4. Drugs causing hypercalcemia**Examples of drugs causing hypercalcemia**

- Excessive calcium intake especially calcium carbonate
- Excessive Vitamin D intake
- Excessive vitamin A intake
- Thyroxin
- Diuretics (Thiazide diuretics e.g. hydrochlorothiazide)

6.5.5. Drugs causing hyperkalemia**Examples of drugs causing hyperkalemia**

- Angiotensin-converting enzyme inhibitors e.g. captopril, enalapril
- Angiotensin II blockers e.g. losartan
- B-blockers
- Heparin
- NSAIDs
- Excessive potassium intake

APPENDIX: Drugs List by Therapeutic Classification
CENTRAL NERVOUS SYSTEM (CNS) DRUGS

Class	Sub-class	Drugs	Comment
General anesthetic	Inhalation	• Desflurane • Halothane	Used for anesthesia prior to general surgery
	Injectable	• Ketamine • Propofol	
Opiates / opioid analgesics		• Morphine • Codeine • Fentanyl • Tramadol	Opiate agonists used for severe pain
Simple analgesics	Alone	Acetaminophen (paracetamol)	Non-opioid, Not antiinflammatory. Used for mild pain and fever. They don't cause addiction
	+ caffeine	Paracetamol + caffeine	
	+ caffeine + codeine	Paracetamol + caffeine + codeine	Contains opioid drugs
NSAIDs (Non-steroidal Anti-inflammatory drugs) Used for mild-moderate pain and	Non-selective	• Aspirin (≥ 300 mg) • Ibuprofen • Ketoprofen • Diclofenac sodium/ potassium • Piroxicam • Indomethacin	Analgesic/anti-inflammatory (used for moderate pain e and mild-moderate inflammation such as arthritis. Act by inhibition prostaglandin synthesis in brain and also periphery. More side effect on stomach and kidney
	Selective COXII inhibitor	• Meloxicam • Celecoxib • Rofacoxib	Less side effects on stomach
Anti-migraine	Alpha-agonist	• Ergotamine + caffeine •	Alpha-1 receptor agonist
	Serotonin agonists	Sumatriptan , Rizatriptan, Zolmitriptan	Selective serotonin receptors antagonists
	Others	• Topiramate	May enhance GABA

Note : **Antipyretics** used for fever : **paracetamol , aspirin , ibuprofen**

CENTRAL NERVOUS SYSTEM (CNS) DRUGS

Class	Sub-class	Drugs	Comment
Antianxiety, sedative, hypnotics (Used in anxiety, stress, insomnia)	Benzodiazepines	• Alprazolam • Bromazepam • Chlordiazepoxide • Diazepam • Lorazepam • Clonazepam	Agonist of BZD part of GABA receptor
	Barbiturates	• Phenobarbitone	Agonist at Barbiturate part of GABA receptor
	Azapirones	• Buspirone	Serotonin agonist
	Others	• Zolpidem	Other mechanism
Antipsychotic (Used to treat psychotic disorders e.g. schizophrenia)	First-generation	• Haloperidol • Chlorpromazine • Thioridazine • Fluphenazine • Promethazine • Thiothixene • Flupenthixol	Dopamine antagonists
	Second-generation	• Clozapine • Olanzapine • Risperidone •	Inhibit serotonin, Noradrenaline, Ach
Antidepressants (used to treat depression)	SSRIs	• Fluoxetine • Paroxetine • Citalopram • Fluvoxamine • Sertraline	Serotonin selective reuptake inhibitors
	SNRIs	• Duloxetine • Venlafaxine	Inhibit serotonin and Noradrenaline reuptake
	Tricyclic	• Amitriptyline • Nortriptyline • Imipramine • Clomipramine	Inhibit serotonin and Noradrenaline reuptake; inhibit beta receptors
	Tetracyclic	• Amoxapine • Maprotiline • Mianserin • Mirtazapine	Inhibit serotonin and Noradrenaline reuptake; inhibit Ach receptors
	Others	• Buspirone	serotonin receptors agonist

CENTRAL NERVOUS SYSTEM (CNS) DRUGS

Class	Sub-class	Drugs	Comment
Antiepileptics/ anticonvulsants (used for seizure and epilepsy)	Barbiturates	• Phenobarbitone	Agonist at Barbiturate part of GABA receptor
	Benzodiazepines	• Lorazepam • Clonazepam • Diazepam • Clobazam • Midazolam • Clorazepate	Agonist at BZD part of GABA receptor.
	Hydantoins	• Phenytoin	Most common anti-epileptics
	Others	• Carbamazepine • Valproic acid, sodium valproate • Lamotrigine • Levetiracetam • Gabapentin • Pregabalin • Topiramate • Tiagabine	
Anti-parkinsonism		• Levodopa + carbidopa • Bromocriptine	Used for parkinsonism

CARDIOVASCULAR SYSTEM (CVS) DRUGS

Class	Sub-class	Drugs	Comment
Antihypertensives (Used to treat hypertension)	Diuretics (used in hypertension, Edema and Congestive heart failure)	<u>Loop diuretics</u> • Furosemide • Torsemide	Increase excretion of water and electrolytes (Na+ , Ca+2 , Mg+2 , K+) in urine.
		<u>Thiazide-diuretics</u> • Hydrochlorothiazide	Increase excretion of water and electrolytes (Na+ , Mg+2 , K+) in urine.
		<u>Thiazide-like diuretics</u> Indapamide	Increase excretion of water and electrolytes (Na+ , Mg+2 , K+) in urine
		<u>Potassium -sparing diuretics</u> • Spironolactone • Amiloride	Increase excretion of water and electrolytes (Na+ , Mg+2 , Ca+2 +) in urine
	Centrally acting	• Methyldopa	Alpha agonist in brain. Used in hypertension of pregnancy
	ACE inhibitor	• Captopril • Enalapril • Lisinopril • Ramipril	Angiotensin converting enzyme inhibitor.
	Angiotensin II blockers (ARBs)	• Losartan • Candesartan • Valsartan	Angiotensin II receptors blockers . Can be also used in angina, MI
	Renin inhibitor	• Aliskiren	
	B-blockers	<u>Non-selective (B1 and B2) blockers</u> • Propranolol • Nadolol • Timolol <u>Selective B1 blockers</u> • Atenolol • Acebutolol	Beta blockers. Also used for angina, MI and cardiac dysrhythmia

		• Bisoprolol • Metoprolol	
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(CARDIOVASCULAR SYSTEM) (CVS) DRUGS

Class	Sub-class	Drugs	Comment
Antihypertensive (Used to treat hypertension)	Alpha blockers	• Doxazosin	Also used in benign prostatic hypertrophy
	Mixed Beta and alpha blockers	• Carvedilol	Used also for CHF and angina
	CCB (Calcium channel blockers)	• Amlodipine • Felodipine • Nifedipine • Diltiazem • Verapamil	Can be also used in angina
	Direct vasodilators	• Hydralazine	Used in severe hypertension
Drugs for CHF (congestive heart failure)	Inotropic	• Digoxin • Dopamine • Dobutamine	Cardiac stimulant
	Others	• Diuretics • ACE inhibitors & ARBs • Some B-blockers (bisoprolol, carvedilol)	Other mechanisms or adjuvants to inotropics
Drugs for (CAD) Coronary Artery Diseases) (Angina, MI (Myocardial infarction)	Nitrates	• Nitroglycerin (glyceryl trinitrate) • Isosorbide dinitrate	Sublingual (for acute case), oral (for prophylaxis)
	Antiplatelets	• Aspirin (75-300 mg) • Clopidogrel • Ticlopidine • Dipyridamole • Tirofiban	Prophylaxis, post syndrome
	Hypolipidemic drugs	<u>Statins</u> Atorvastatin Pravastatin Simvastatin Rosuvastatin lovastatin <u>Others</u> Fenofibrate Cholestyramine Gemfibrozil Omega 3 fatty acids Niacin (nicotinamide = Vit. B 3)	Decrease level of lipid in blood

(CARDIOVASCULAR SYSTEM) (CVS) DRUGS

Class	Sub-class	Drugs	Comment
Antiarrhythmics (Antidysrhythmics) Used in cardiac dysrhythmia	Class I	• Disopyramide • Lidocaine (i.v.) • Flecainide • Propafenone	Block Na ⁺ channels
	Class II	• Propranolol, acebutolol	Beta blockers
	Class III	• Amiodarone	Block Na ⁺ , K ⁺ , and Ca ²⁺ channels
	Class IV	• Verapamil, Diltiazem	CCB
	Class V	• Adenosine	Adenosine 2 receptor agonist

DRUGS FOR BLOOD DISORDERS

Class	Sub-class	Drugs	Comment
Antihemorrhagics (Hemostatics) used to treat bleeding	Only in case of vitamin K deficiency	• Vitamin K	Only if bleeding is caused by Vitamin K deficiency
	For general cases of bleeding	• Tranexmic acid • Aprotinin •	Inhibit fibrinolysis
		• Etamsylate	Increase capillaries endothelium and increase platelet aggregation
	For visceral bleeding	• Octreotide	
	Topical	• Tranexmic acid • Etamsylate	
Anticoagulants (used to inhibit over-coagulation)	Injections	• Heparin • Enoxaparin	Inactivate clotting factors
	Oral	• Warfarin	Inhibits Vitamin K synthesis
Antanemics	See “ dietary supplements” iron, folic acid, vitamin B12		
Iron poisoning		• Desferoxamine	

RESPIRATORY SYSTEM DRUGS

Class	Sub-class	Drugs	Comment
Drugs for bronchial asthma (Bronchodilators)	Inhalation	<u>B2- agonists</u> • Salbutamol (albuterol) • Salmeterol • Formeterol <u>Corticosteroids</u> • Budesonide • Fluticasone • beclomethasone <u>Anti-cholinergics</u> • Ipratropium • Tiotropium	are used as aerosol or inhalation in acute cases
	Oral	<u>B2- agonist</u> • Salbutamol (albuterol) • Terbutaline <u>Beta and alpha agonist</u> • Ephedrine <u>Corticosteroids</u> • Prednisolone • Methylprednisolone • Betamethasone • Dexamethasone <u>Leucotriene receptor blockers</u> • Montelukast • Zafirlukast <u>Methylxanthines</u> • Theophylline • Aminophylline	Used for prophylaxis
	Injections	• Corticosteroids • Aminophylline	Used for acute cases
Drugs for cough	For dry cough	• Dextromethorphan • Codeine • Oxomemazine • Butamirate • Diphenhydramine	Dry cough : no sputum
	For wet cough	• Guaifenesin • Bromohexine • Ambroxol • Ammonium chloride	Wet cough (cough with sputum)

RESPIRATORY SYSTEM DRUGS

Class	Sub-class	Drugs	Comment
Drugs for cold	Antihistamines	<u>Sedative antihistamines (1st</u>	Antihistamines

	(Antallergics)	<u>Generation)</u> • Chlorpheniramine • Clemastine • Cyproheptadine • Diphenhydramine • Hydroxyzine • Triprolidine • Dimetidine <u>Less Sedative antihistamines (2nd Generation)</u> • Cetirizine • Levocetirizine • Loratadine • Desloratadine • Fexofenadine	can be also used for allergy cases
	Nasal decongestants	<u>Oral</u> • Pseudoephedrine • Ephedrine • Phenylephrine <u>Nasal</u> • Oxymetazoline • Xylometazoline • Naphazoline • Phenylephrine	Alpha 1 agonists
Many products for cold may also contain analgesic/antipyretics e.g. paracetamol			

DRUGS FOR MUSCLES/JOINTS DISORDERS (myalgia, gout and arthritis)

Class	Sub-class	Drugs	Comment
Mild-moderate skeletal pain and inflammation (myalgia, arthritis)	NSAIDS	<u>See previous</u>	
Anti-gout	Gout	• Allopurinol • Colchicine • Febuxostat	Used also in other cases of hyperuricemia
For severe rheumatoid arthritis	Potent immunosuppressants	• Cyclosporine • Methotrexate • Hydroxychloroquine	

GASTROINTESTINAL (GIT) DRUGS

Class	Sub-class	Drugs	Comment
Drugs for hyperacidity and peptic ulcer disease (PUD) and GERD (gastroesophageal reflux disease)	Neutralizing Antacids	• Sodium bicarbonate • Potassium bicarbonate • Magnesium hydroxide, magnesium trisilicate • Calcium carbonate	Used in mild cases
	H2-blockers	• Famotidine • Nizatidine	Used in moderate-severe cases
	Proton pump inhibitors (PPIs)	• Omeprazole • Lansoprazole • Esomeprazole • Pantoprazole • Rabiprazole	Used in severe cases
Antiemetics (used in vomiting, nausea and motion sickness)		<u>Anti-motion sickness</u> • Dimenhydrinate • Diphenhydramine • Meclizine <u>Used for vomiting</u> • Metoclopramide • Domperidone • Meclizine • Ondansetron • Granisetron • Vitamin B6	Meclizine, vitamin B6, ondansetron can be used in pregnancy
Antispasmodics (used for intestinal colic, biliary colic (with NSAIDs) and uterine colic (pregnant women)		<u>Anti-cholinergics</u> • Hyoscine • Atropine • Dicyclomine • Clinidium <u>Non anti-cholinergic</u> • Drotaverine • Mebeverine	Anti-cholinergics cause dry mouth, visual disturbance and tachycardia
Laxatives (for constipation)		• Castor oil • Paraffin oil • Bisacodyl • Glycerin • Senna • Sodium picosulfate • Sodium phosphate enema	Bisacodyl, castor oil , senna and milk of magnesia cause colic

GASTROINTESTINAL (GIT) DRUGS

Class	Sub-class	Drugs	Comment
Antidiarrheals		• Loperamide • Atropine + diphenoxylate • Bismuth subsalicylate • Psyllium	They reduce diarrhea but don't treat the cause)
For indigestion and flatulence		• Digestive enzymes (lipase, amylase, protease) • Medicated Charcoal • Simethicone , dimethicone • Sodium bicarbonate	

ENDOCRINOLOGY (HORMONES & HORMONE-RELATED)DRUGS

Class	Sub-class	Drugs	Comment
Growth hormone (GH) and related drugs	Natural	• Growth hormone	GH deficiency and as anabolic agent
	GH analogue	• Somatotropin	
GH inhibitor (inhibit GH secretion)		• Octreotide • Lanreotide	Somatostatin analogue (used for acromegaly and visceral bleeding)
Antidiabetics (used for diabetes mellitus to decrease hyperglycemia)	Insulin	• Insulin (short acting; moderate acting; long acting)	They may cause serious hypoglycemia .
	Oral hypoglycemics	<u>Sulfonylureas</u> • Glibenclamide (Glyburide) • Glipizide • Glimepiride <u>Alpha glucosidase inhibitors</u> • Acarbose <u>Biguanides</u> • Metformin <u>Meglitinides</u> • Rapaglinide <u>Thiazolidinediones</u> • Pioglitazone • Rosiglitazone <u>Glucagon-like peptides</u> • Lixisenatide • Semaglutide • Liraglutide	
Thyroid disorders drugs	Thyroxin	• Levothyroxine (T₄ synthetic) • Liothyronine (T₃ synthetic)	Used for Hypothyroidism
	Antithyroid drugs	• Potassium iodide • Methimazole • Propylthiouracil	Used for hyperthyroidism, Grave's diseases

ENDOCRINOLOGY (HORMONES & HORMONE-RELATED)DRUGS

Class	Sub-class	Drugs	Comment
Male sex hormone and related drugs (Androgens)	Testosterones	• Testosterone • Methytestosterone	Used in testosterone deficiency in males
	Anabolic steroids	• Oxandrolone	Used for weight gain
	Other androgens	• Danazol	Used for endometriosis and fibrocystic breast in women
Anti-androgens	block testosterone receptors	• Flutamide • Nilutamide • Apalutamide •	Used for Prostate cancer in males
	inhibit testosterone synthesis /	• abiraterone	
Female sex hormones		<u>ESTROGENS</u> • Estradiol • Estradiol acetate • Ethinyl estradiol • Conjugated estrogens • Estropipate <u>PROGESTINS</u> <u>Natural</u> • Progesterone <u>Synthetic</u> • Hydroxyprogesterone • Norgestrel • Levonorgestrel • Etonogestrel • Norethindrone • Medroxyprogesterone • Megestrol • Norgestimate <u>Estrogen + progestin combination</u> • Estradiol + Levonorgestrel • Estradiol + norgestimate • Estradiol + norethindrone	Used as contraceptives and for other women disorders)

ENDOCRINOLOGY (HORMONES & HORMONE-RELATED)DRUGS

Class	Sub-class	Drugs	Comment
Anti-estrogen	To induce ovulation	• Tamoxifen • Clomiphene	
	For breast cancer	• Tamoxifen • Clomiphene • Torimefene • Fluvestrant	
Gonadotropins (used for infertility)		• Human chorionic gonadotropin	stimulate follicular maturation and ovulation in females and spermatogenesis in males
Prolactin inhibitor		• Bromocriptine	Dopamine agonist that inhibit prolactin secretion. Can be used in : parkinsonism, infertility and other types of hyperprolactinemia
• GNRH inhibitors		• Leuprolide • Goserelin	inhibit gonadotropin releasing hormone. Used anticancers

Note : Steroids : (androgens), (estrogens), (progestins), cortocosteroids

ANTI-INFECTIVE DRUGS

Class	Sub-class	Drugs	Comment
Antibacterials (B-lactams)	Penicillins	<u>Narrow-spectrum (G +ve)</u> • Crystalline penicillin (Penicillin (G) • Procaine penicillin G • Benzathine penicillin G • Phenoxymethyl penicillin (Penicillin V) <u>Broad-spectrum (but not resistant to betalactmase)</u> • Ampicillin • Amoxicillin <u>Broad-spectrum (Extended spectrum)</u> • Amoxicillin + clavulenate • Piperacillin + tazobactam • Ampicillin + sulbactam	The first ever discovered antibiotics
	Cephalosporins	<u>1st Generation (Mainly G+ve)</u> • Cefazolin • Cephalexin • Cefadroxyl • Cephadrine <u>2nd Generation (Broad spectrum)</u> • Cefuroxime • Cefaclor • Cefprozil • Cefotetan <u>3rd Generation (Extended spectrum)</u> • Cefixime • Ceftriaxone • Cefotaxime • Ceftizidime • Cefoperazone <u>4th Generation (Extended spectrum)</u> • Cefepime	Very commonly used
	Carbapenemes	• Imipenem • Etrapanem	Very Extended spectrum
	Monobactam	• Aztronam	G-ve

Broad-spectrum (active against G+ve and G-ve); **Extended-spectrum** (G+ve, G-ve including some resistant bacteria). **Antibiotic** is any anti-infective or anti-cancer derived from microorganism .

ANTI-INFECTIVE DRUGS

Class	Sub-class	Drugs	Comment
Other antibacterials	Oxazolidinones	• Linezolid	G+ve resistant bacteria
	Macrolides	• Erythromycin • Azithromycin • Clarithromycin • Roxithromycin	Erythromycin act mainly on G+ve. Other macrolides act on G+ve, G-ve
	Aminoglycosides	• Gentamicin • Amikacin	Act mainly on G-ve
	Tetracyclines	• Tetracycline • Doxycycline • Minocycline	Bacteriostatic and Broad-spectrum
	Lincosamides	• Lincomycin (Injection) • Clindamycin (oral /injection)	Narrow G+ve
	Glycopeptides	• Vancomycin (injection)	G+ve resistant bacteria; also for Clostridium
	Fluroquinolones	• Ciprofloxacin • Ofloxacin • Norfloxacin • Lomefloxacin' • Levofloxacin	Antibacterial but Not antibiotic. Broad spectrum (G+ve, G-ve)
	Sulfonamides	• Sulfamethoxazole + trimethoprim) also called (Cotrimoxazole) (oral) • Sulfacetamide (ophthalmic)	Antibacterial but Not antibiotic. Cotrimoxazole used mainly for bacterial dysentery and UTI
	Nitroimidazoles	• Metronidazole • Tinidazole • Secnidazole	Antibacterial but Not antibiotic Effective only for anaerobic bacteria
Anti-TB (used for Mycobacterium tuberculosis)		• Rifampicin • Isoniazide • Streptomycin • Ethambutol	Usually used as combination
Anti-leprosy drugs (used for Mycobacterium leprosy)		• Dapsone • Rifampicin • Clofazimine	

ANTI-INFECTIVE DRUGS

Class	Sub-class	Drugs	Comment
Antifungal		<u>Weak</u> • Nystatin (oral /topical) • Griseofulvin (oral) <u>Moderate</u> • Miconazole (oral/topical) • Clotrimazole (topical) • Itraconazole (oral) • Ketoconazole (oral/topical) <u>Potent</u> • Tolnaftate (Topical) • Terbinafine (oral /topical) • Flucytosine (oral)	Used for fungal infections - Skin - nail - vagina
Antivirals	For Herpes virus	• Acyclovir	Oral / topical
	For CMV (cytomegalovirus)	• Ganciclovir	Injection
	For influenza virus	• Amantadine • Rimantadine • Oseltamivir • Zanamivir	Oral
	For HIV (AIDS)	<u>Protease inhibitors</u> • Ritonavir • Saquinavir • Atazanavir • Indinavir • Lopinavir • Indinavir <u>Non-protease inhibitors</u> • Zidovudine • Abacavir • Dedanosine • Lamivudine • Didanosine <u>Monoclonal antibodies</u> • Ibalizumab	They are also called antiretrovirals
	For Hepatitis C virus	• Interferon alfa 2-b • Ribavirin	Interferon : injections Ribavirin: oral
	For Hepatitis B virus	• Lamivudine • Adefovir • Entecavir	Oral

ANTI-INFECTIVE DRUGS

Class	Sub-class	Drugs	Comment
Antiprotozoal	Antamoebics Antigiardials	• Metronidazole • Tinidazole • Secnidazole	Used for protozoa (amoeba , giardia , trichomonas) and anaerobic bacteria
	Antimalarials	• Chloroquine • Hydroxychloroquine • Sulfadoxine + pyrimethamine • lumifantrene • Artemether • Artesunate • Quinine • Primaquine • Atovaquone • Proguanil	Used for malaria infection
	Anti-leishmania	• Sodium stibogluconate	Injection
	Anti-toxoplasmosis	• Spiramycin • Sulfonamide + pyrimethamine	Used for toxoplasmosis
Anthelmintics	Common worms	• Mebendazole • Albendazole • Pyrantel pamoate	Common worms e.g. Pin worm, Ascaris, Trichuris , ancylostoma
	Tape worms	• Niclosamide • Praziquantel	Oral
	Schistosoma (Bilharzia)	• Praziquantel	
	River blindness worms, Filaria worms	• Ivermectin • Diethylcarbamazine	

ANTICANCERS (ANTINEOPLASTICS) DRUGS

Class	Sub-class	Drugs	Comment
Cytotoxic drugs	Alkylating agents	• Carmustine • Cisplatin • Cyclophosphamide • Busulfan • Dacarbazine	Cytotoxic drugs are non-targeting drugs used to treat cancer. They cause more severe adverse effects than other antineoplastics
	Antimetabolites (antifolates)	• Methotrexate • 6-Mercaptopurine • 6-Thioguanine • Fluorouracil • Cytarabine • Hydroxycarbamide	
	Anthracyclines	• Doxorubicin • Daunorubicin • Valrubicin	
	Plant alkaloids	• Vinblastine • Vincristine	
	Antibiotics	• Bleomycin • Dactinomycin • Mitomycin	
	Others	• Cabazitaxel • Paclitaxel • Ixabepilone • Irinotecan • Topotecan • Azacitidine • Decitabine	
Targeted	Monoclonal antibodies (Mab)	• Ofatumumab • Obinutuzumab • Ibritumomab • Pertuzumab	Targeted antineoplastics are anticancer less That cause less adverse effects than cytotoxics
	Aromatase inhibitors	• Anastrozole • Exemestane • Letrozole	
	Others	• Brigatinib • Ceritinib • Lorlatinib • Regorafenib • Afatinib • Dacomitinib • Osimertinib	

ANTICANCERS (ANTINEOPLASTICS) DRUGS

Class	Sub-class	Drugs	Comment
Hormones and hormone antagonists	Sex hormones	• Androgens • Estrogens • Progestins	Androgens for breast cancer. Estrogens for prostate cancer
	Antiandrogens	• Flutamide • Nilutamide • Apalutamide • abiraterone	Used for prostate cancer
	Antiestrogens	• Tamoxifen • Clomiphene • Torimefene • Fluvestrant	Used for breast cancer
	GNRH inhibitors	• Leuprolide • Goserelin	inhibit gonadotropin releasing hormone

IMMUNOSUPPRESSANT DRUGS

Class	Sub-class	Drugs	Comment
Potent immunosuppressants		• Cyclosporine • Abatacept • Tacrolimus • Azathioprine • Infliximab • Methotrexate • Hydroxychloroquine	Potent immunosuppressants Used in autoimmune diseases (e.g. rheumatoid arthritis , psoriasis) and organ transplantation
Corticosteroids	Oral/ injection	* <u>Mineralocorticosteroids</u> • Fludrocortisone * <u>Glucocorticosteroids</u> <u>Potent-antiinflammatory</u> • Dexamethasone • Betamethasone • Triamcinolone <u>Moderate- antiinflammatory</u> • Prednisone • Prednisolone • Methylprednisolone <u>Weak-antiinflammatory</u> • Hydrocortisone	*Mineralocorticosteroids have no anti-inflammatory effect. They are used in adrenocortical deficiency (Addison disease). They have * <u>Glucocorticosteroids</u> have anti-inflammatory effect. They have also anti-allergic and immunosuppressant effect.
	Topical	Topical corticosteroids <u>Very potent</u> • Clobetasol <u>Potent</u> • Betametasone dipropionate • Fluticasone <u>Medium</u> • Betamethasone valerate • Flucinolone <u>Weak</u> Hydrocortisone	

- Mineralocorticoids : have mineralocorticoid activity (water retention, hypomagnesemia, hypophosphatemia, hypokalemia, hypernatremia) causing edema and increase blood pressure
- Glucocorticoids : Potent-antiinflammatory drugs and moderate-antiinflammatory drugs (weak mineralocorticoid activity); Weak-antiinflammatory (moderate mineralocorticoid activity)

NUTRIENTS (CARBOHYDRATES, VITAMINS AND MINERALS)
Used as dietary supplements and also in deficiency of vitamins /
minerals.

Class	Sub-class	Drugs	Comment
Carbohydrates	Dextrose (glucose)	• Oral : 40 % • Injection (I.V.) - Infusion : dextrose 5% - Injection : dextrose 40 %	Used in hypoglycemia
Carbohydrates + Minerals		• Oral : Oral rehydration salt ORS (Sodium 75 mmole/L) . Also contains chloride (65 mmol/L), potassium (20 mmole/L), bicarbonate (30 mmole/L) and glucose (111 mmole/L) • Injection (I.V.) infusion DNS (GNS): dextrose 5 % + sodium chloride 0.9 %	Used in dehydration
Minerals	Sodium	• Oral : ORS • I.V.: - Normal saline infusion 500 ml 0.9% sodium chloride. - Hypotonic 0.45% injection - Ringer-lactate infusion 500 ml (NaCl 0.6% , KCl 0.03%, CaCl₂ 0.02 % , Na. lactate 0.3 %)	Used in Dehydration
	Calcium	• Oral: Calcium carbonate (500 mg calcium), OR calcium gluconate (500 mg calcium), OR calcium lactate gluconate + calcium carbonate (500 mg calcium) • Injection (I.V.): Calcium chloride, OR calcium gluconate (10 %)	Used in hypocalcaemia
	Potassium	• Oral : potassium chloride 600 mg • Injection (I.V.): potassium 10 %	Used in hypokalemia
	Iron	<u>Oral</u> • Ferrous sulphate 200 mg (65 mg elemental iron) • Ferrous fumarate 150 mg (50 mg elemental iron) gluconate • Ferric gluconate 300 mg (45 mg elemental iron) <u>Injection</u> • I.M: Iron (III) hydroxide complex (100 mg elemental iron) • I.V / I.M.: Iron –dextran complex (100 mg elemental iron) • I.V.: Iron sucrose (100 mg elemental iron)	Used in iron deficiency anemia

NUTRIENTS (CARBOHYDRATES, VITAMINS AND MINERALS)

Class	Sub-class	Drugs	Comment
Vitamins	Vitamin A	• Oral : : 50000 IU (15 mg)	Natural Vitamin A (B-carotene)
	Vitamin K (Also in hypoprothrombinemia, Reversal warfarin side effects)	• Oral : 1 mg • Injection: 1 mg i	Phytonadione
	Vitamin D (used in vitamin D deficiency and also in osteoporosis, hyperparathyroidism, Rickets_	• Oral 400 IU (10 mcg), 2000 IU (50 mcg) • Injection 100,000 IU (2.5 mg) Intramuscular	Cholecalciferol
		• Oral : 0.25 mcg	Alfacalcidol
	Vitamin E	• Oral : 200 IU, 400 IU	Tocopherol
	Vitamin C	• Oral : Vitamin C (ascorbic acid) 500 mg tablet, 1000 mg effervescent tablets, effervescent granules • Injection 500 mg	Also used to stimulate immunity
	Vitamin B6	• Oral tablet 250 mg • Injection 250 mg	Used for Vitamin B6 deficiency and also for vomiting vitamin B 6 d
	Folic Acid (Vitamin B9)	• Oral : 5 mg • Injection : 5 mg	Used in megaloblastic anemia due to folate-deficiency
	Vitamin B12	• Oral : Cyanocobalmin OR methycobalamin 1 mg tablet • Injection : Cyanocobalamine OR methycobalamin 1 mg	For megaloblastic anemia (pernicious anemia) due to vitamin B12 deficiency

Note : A combination of folic acid + vitamin B12 can be used in megaloblastic anemia due to deficiency in both two vitamins

NUTRIENTS (CARBOHYDRATES, VITAMINS AND MINERALS)

Class	Sub-class	Drugs	Comment
Vitamins	Vitamin B1, B6, B12	• Oral : Vitamin B1 (100 mg) + vitamin B6 (200 mg) + vitamin B12(200 mcg) • Injection (I.M): Vitamin B1 (100 mg) + vitamin B6 (100 mg) + vitamin B12(1000 mcg)	for vitamin B1, B6 and B12 deficiency including neuritis
	Vitamin B complex	• Oral : B1 (10 mg) + B2 (10 mg) + B6 (3 mg) + B 12 (10 mcg) + B7 biotin (0.15 mg) + B3 (niacin; nicotinamide (50 mg) • Injection : B1 (10 mg), B2 (10 mg) + B6 (3 mg) + B 12 (10 mcg) + biotin (0.15 mg) + B3 (Niacin ; nicotinamide (40 mg) + B5 (pantothenic acid ; dexpantenol (6 mg); B6 (4 mg)	Used in peripheral neuritis
	Multivitamins	Combination of vitamins	Contain no minerals.
Minerals + Vitamins	Vitamin C + calcium	Oral : Vitamin C (1 g) + calcium salt (260 mg calcium)	Stimulate immunity
	Vitamin C + zinc	Oral : vitamin C (1 g) + zinc 10 mg	
	Vitamin D + calcium	Oral : Vitamin D (10 mcg) + Calcium salt (500 mg calcium)	Increase effect of calcium by vitamin D
	Vitamin A + Vitamin E + zinc	Oral : Vitamin A (50000 IU) , zinc gluconate 175 mg, Vitamin E (100 mg)	Used in skin/hair/ eye disorders
	Multivitamin + minerals	Combination of vitamins + minerals	Multivitamins + minerals